THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

V. CASAÑA-GINER

2020 *Mathematics Subject Classification.* Primary 11N37; Secondary 11M26, 60G70.

Key words and phrases. Mertens function, Möbius function, excursions, Riemann hypothesis, Kaczorowski–Pintz oscillation, lacunary sequences, Dyck paths, Catalan tails, extreme values.

Abstract. We decompose the Mertens function $M(x)=\sum_{n\le x}\mu(n)$ into its excursions between zero runs and study a regularized triangular skeleton of these excursions, its area curves, and exact quadratic and correlation identities inside each completed excursion.

For any ordered family of disjoint integer intervals $(x_0,x_1]$, elementary counting shows that the members starting at most X with $x_1>x_0^A$ $(A>1)$ are $O_A(\log\log X)$, those with $x_1>cx_0^2$ are $O_c(\log\log X)$, those with $x_1>x_0\psi(x_0)$ are $o(\log X)$ whenever $\psi\to\infty$, and those with $x_1>\lambda x_0$ are at most $\log X/\log\lambda+O(1)$. Combined with the Kaczorowski–Pintz oscillation theorem, this gives a horizon-localization principle: if the excursions starting at most X that belong to an exceptional family or cross a horizon $\mathfrak{h}$ number at most $(\kappa+o(1))\log X$ with $\kappa<\gamma_1/\pi$, then the Riemann hypothesis is equivalent to Littlewood's bound $|M(t)|\ll_\varepsilon t^{1/2+\varepsilon}$ restricted to $x_0<t\le\min\{x_1,\mathfrak{h}(x_0)\}$ on the remaining excursions. The horizon can be lowered from x_0^2 through cx_0^2 and $x_0\psi(x_0)$ to λx_0 for every $\lambda>\lambda_*:=e^{\pi/\gamma_1}$, the sharp threshold of this counting mechanism. The same input shows that late excursions end at $x_1=x_0^{1+o(1)}$, so that their length satisfies $\ell=x_1-x_0\le x_0^{1+o(1)}$. On the resulting window every $\theta>\theta_*:=\frac12 e^{\pi/\gamma_1}$ is an admissible prefix-dependency cutoff: with $M(u)=0$ and $u<t\le\lambda u<2u$,

$$M(t)=-\sum_{d\le\lambda u/2}\mu(d)\Big(\Big\lfloor\frac{t}{d}\Big\rfloor-\Big\lfloor\frac{u}{d}\Big\rfloor\Big).$$

After deleting zero increments, a completed excursion is a primitive Dyck path on its fixed support mask. Under the uniform primitive Catalan reference law on that mask, a value $|M(t)|\ge t^{1/2+\varepsilon}$ in an excursion with $x_1\le\lambda x_0$ has probability at most $x_0^{O(1)}e^{-x_0^{2\varepsilon}/2}$, and polynomial lower bounds for the complete-excursion or open-prefix tail weights imply $|M(t)|\ll\sqrt{t\log t}$ on the corresponding ranges.

The one-step recursion concentrates, for every fixed $K\ge2$, all dependence above $\lambda u/(K+1)$ in the $K-1$ channels $M(t/k)-M(u/k)$, $2\le k\le K$, and a persistence lemma quantifies how a large value must be carried by one of them or by the deeper remainder. A multiplicative-layer filtration shows that for every fixed depth k the first k Möbius layers determine the $(k+1)$-st layer of the future window; the determined positions have density zero, while their signed contribution is of order $u(\log\log u)^k/\log u$. Thus substantial cancellation with the remaining layers is necessary to reach the RH scale; no assertion on the number of layers required is made.

Inside a completed excursion, signed lag averages of two-point correlations are forced by the return to zero, while the weighted identity $Q=-\sum_h hC_h$ holds globally at every zero of M with exact cancellation of the cross-excursion terms; this yields an RH criterion in terms of h-weighted correlations at zeros of M, showing that RH is a statement of cancellation between lags. Combining the Fourier-uniform estimate of Matomäki–Teräväinen with Parseval, we obtain uniform two-point Cesàro cancellation in every interval $(u,u+L]$ in the proved range $u^{3/5+\eta}\le L\le u$, averaged over lags $h\le R$ with $R\ge L(\log u)^{-2+\delta}$ for every fixed $0<\delta<1$; it applies to the linearly localized Kaczorowski–Pintz excursions with $v\le\lambda u$ and fixed $\lambda<2$, provided $\Theta>3/5$. A separate Fejér reduction shows that, for complete excursions and $R=o(\sqrt{L})$, the weaker target $F_R=o(RL)$ is equivalent to a diagonal local-variance law for increments of M; this target may in principle extend below the 3/5 Fourier threshold.

Theorems 2.1–2.2 of Pintz [18] on the logarithmic mean and late maximal order of the Mertens function give a further consequence of the excursion decomposition. Writing $\Theta=\sup\Re\rho$, the cumulative absolute excursion area, the quadratic mass, and the chord-energy functional introduced here satisfy

$$\sum_{n\le X}|M(n)|=X^{1+\Theta+o(1)},\qquad\sum_{n\le X}M(n)^2=X^{1+2\Theta+o(1)},\qquad\mathcal{E}(X)=X^{1+\Theta+o(1)}.$$

Moreover, if $D_I(s)=\sum_{n\in I}\mu(n)n^{-s}$ denotes the Dirichlet block of a completed excursion, then the block series $\sum_I D_I(s)$ has abscissa of absolute convergence exactly Θ and represents $1/\zeta(s)$ throughout $\Re s>\Theta$. At zeros v of M this also yields $-\sum_{h<v}hC_h^{[v]}=v^{1+2\Theta+o(1)}$; combined with averaged Chowla, all lags $h\le v^\alpha$ are negligible relative to this weighted mass for every fixed $\alpha<\Theta$.

Under a uniform sub-Weibull tail hypothesis for a normalized excursion statistic, the counting bounds give extreme-value scales by a union bound and a Borel–Cantelli argument. We also give an exact bilinear representation of M on the square range of an excursion, an exact Mellin decomposition of $1/\zeta$ into completed excursion blocks, and asymptotic decompositions of M into smooth and large-prime parts.

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1. Introduction

The object of this paper is the Mertens function M, decomposed into excursions between zero runs (Section 3). Two kinds of information about M are separated throughout: what is forced by the ordering and local geometry of the excursions alone, and what requires arithmetic input.

Main results. The central results are in Section 9.

Geometric counts. For any ordered family of disjoint integer intervals $(x_0, x_1]$, the members with $x_1 > x_0^A$ $(A > 1)$ starting below X are $O_A(\log\log X)$, those with $x_1 > cx_0^2$ are $O_c(\log\log X)$, those with $x_1 > x_0\psi(x_0)$ are $o(\log X)$ whenever $\psi \to \infty$, and those with $x_1 > \lambda x_0$ are at most $\log X/\log\lambda + O(1)$ (Lemmas 9.1, 9.2, 9.3, 9.4). No arithmetic enters.

Horizon localization. By Kaczorowski–Pintz [9], if $\Theta > 1/2$ then at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting below X contain values of size $t^{\Theta-\eta}$. Theorem 9.11 turns every geometric count below this threshold into an equivalent form of RH: for any horizon h whose crossing family has at most $(\kappa + o(1))\log X$ members with $\kappa < \gamma_1/\pi$,

$$\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad \text{for } x_0 < t \le \min\{x_1, \mathfrak{h}(x_0)\}.$$

In particular (Corollary 9.13)

$$\text{RH} \iff \text{Littlewood's bound holds on } x_0 < t \le \min\{x_1, \lambda x_0\}, \qquad \lambda > \lambda_* := e^{\pi/\gamma_1} = 1.2488\ldots,$$

and the square horizon x_0^2 is eroded from above through cx_0^2 and $x_0\psi(x_0)$. If RH fails, a positive proportion (on the scale $\log X$) of the Kaczorowski–Pintz excursions carry their large value in $[x_0 + \frac{1}{2}x_0^{\Theta-\eta}, \lambda x_0]$ (Corollary 9.14). The threshold λ_* is the multiplicative distance between consecutive sign changes of $\cos(\gamma_1 \log x)$ (Remark 9.29). It is the sharp threshold of this counting mechanism, not a proved limit for other localization methods. The local form of the Kaczorowski–Pintz theorem also shows that for every fixed $A > 1$ all late excursions satisfy $x_1 \le x_0^A$ (Corollary 9.8): the endpoints of late Mertens excursions satisfy $x_1 = x_0^{1+o(1)}$, and hence $\ell = x_1 - x_0 \le x_0^{1+o(1)}$; no lower bound on ℓ follows. The fractional-square lemma makes the intermediate erosion explicit:

$$x_0^2 \longrightarrow \frac{x_0^2}{2} \longrightarrow \frac{x_0^2}{4} \longrightarrow \frac{x_0^2}{8} \longrightarrow \cdots \longrightarrow x_0\psi(x_0) \longrightarrow \lambda x_0.$$

For each fixed dyadic level the crossing family is $O(\log\log X)$; the passage to a level depending on x_0 is supplied by the superlinear-horizon lemma, not by a uniform use of the fixed-c estimate.

Kaczorowski–Pintz [9] provide the oscillation-density input. The excursion-counting lemmas, the horizon-localization theorem, the descending-horizon principle, the threshold $\lambda_* = e^{\pi/\gamma_1}$, and the resulting prefix-dependency cutoff $\theta_* = \frac{1}{2}e^{\pi/\gamma_1}$ are developed here from that input.

Linear window. For $u < t \le 2u$ one has $\mu(n) = -\sum_{d|n,\, d\le u} \mu(d)$. After horizon localization, if $u < t \le \lambda u < 2u$, the terms with $d > \lambda u/2$ have no multiple in $(u, t]$, and therefore

$$M(t) = -\sum_{d \le \lambda u/2} \mu(d)\Big(\Big\lfloor \frac{t}{d} \Big\rfloor - \Big\lfloor \frac{u}{d} \Big\rfloor\Big).$$

Thus every $\theta > \theta_* := \frac{1}{2}e^{\pi/\gamma_1}$ is an admissible prefix cutoff for the RH-equivalent excursion window; the convenient choice $\lambda = 5/4$ uses only $d \le 5u/8$. This is a descent of dependency, not of size. More sharply, for every fixed $K \ge 2$ the dependence above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $M(t/k) - M(u/k)$, $2 \le k \le K$; for $K = 2$ and $\lambda = 5/4$, the only channel above $5u/12$ is the increment across $(u/2, t/2]$. The one-step recursion does not by itself give an iterable maximum principle, and a termwise absolute-value contraction gives no nontrivial gain. The exact intra-excursion formulation, the scale cascade, a persistence extraction lemma, and a ballistic counterexample to the relaxed divisor-sum version are given below. A bounded number of multiplicative layers determines only a density-zero set of increments in the localized window, but for every fixed depth k the signed contribution of the determined layers is of order $u(\log\log u)^k/\log u$, far above the RH scale; reaching the RH scale requires substantial cancellation with the remaining layers, and the present arguments do not determine how many layers are needed.

Section 10 connects the excursions with averaged forms of Chowla's conjecture. Signed lag averages of correlations inside a bridge are forced by the return to zero, while the weighted identity $Q_\nu = -\sum_h hC_h^{(\nu)}$ holds globally at every zero of M with exact cancellation of the cross-excursion terms. This gives an RH criterion in terms of h-weighted correlations at zeros of M and shows that RH is a statement of cancellation between lags. Combined with the averaged Chowla theorem of Matomäki–Radziwiłł–Tao, it forces the energy of linear-scale high excursions beyond every admissible small-lag cutoff. Fourier uniformity in all intervals in the range $u^{3/5+\eta} \le L \le u$ extends absolute two-point Cesàro cancellation to that range. Separately, the Fejér identity reduces

a weaker local question on complete excursions to a diagonal variance law for short increments of M; this is not equivalent to absolute Chowla or to localization of the full weighted energy.

Exact logarithmic exponents. Pintz's 2026 theorem on the logarithmic mean and late maximal order implies that the average and late maximal orders of $|M|$ have logarithmic exponent Θ. Transferred to the excursion variables, this gives

$$\sum_{n\le X}|M(n)| = X^{1+\Theta+o(1)}, \qquad \sum_{n\le X}M(n)^2 = X^{1+2\Theta+o(1)}, \qquad \mathcal{E}(X) = X^{1+\Theta+o(1)}.$$

Thus the area and energy quantities recover the rightmost-zero exponent exactly, not merely through RH-equivalent upper bounds. The same input shows that the completed-excursion blocks of $1/\zeta$ have abscissa of absolute block convergence exactly Θ. At a zero v of M the weighted correlation identity therefore has size $v^{1+2\Theta+o(1)}$, and MRT implies that, for each fixed $\alpha<\Theta$, the part from $h\le v^{\alpha}$ is $o(1)$ of the total weighted mass.

Extreme values. If a normalized excursion statistic has a uniform tail $\Pr(Z>z)\le Ce^{-\kappa_0 z^{\varpi}}$, the counting bounds force the maximum over a family with at most $\Lambda(X)$ members starting below X to be of order $(\log\Lambda(X))^{1/\varpi}$ in probability, with an almost sure lim sup constant $\kappa_0^{-1/\varpi}$ (Theorems 9.70 and 9.72). Each level of lacunarity removes one logarithm from the extreme scale: $\sqrt{\log X}$, $\sqrt{\log\log X}$, $\sqrt{\log\log\log X}$.

The remaining sections contain exact identities for the skeleton and its area curves (Sections 3–4), exact correlation identities and sharp bounds inside an excursion (Section 5), the relation with ζ (Section 6), area bounds and global criteria equivalent to RH (Sections 7–8), the exact bilinear form of M on the square range of an excursion and the Mellin mass of an excursion (Section 9), and the decomposition of M by large prime factors, globally and inside an excursion (Sections 11–13). Section 15 records numerical data on the excursion census.

Conventions. ρ runs over nontrivial zeros of ζ, with ordinate $\gamma=\Im\rho$,

$$\Theta=\sup\{\Re\rho\}, \qquad \gamma_1 = 14.134725\ldots$$

is the ordinate of the lowest zero, and RH means $\Theta=1/2$. Littlewood's criterion [21, §14.25] states that RH holds if and only if $M(x)\ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon>0$. Statements labelled *Heuristic* are not used in any proof.

2. Notation and variables

Global symbols keep the meanings listed below. Local dummy variables may be reused inside a single proof only when their scope is explicit. In particular, σ_ν is reserved for excursion orientation, whereas the complex variable is always written $s=\varsigma+i\tau$. Constants denoted by c, C (with unspecified subscripts) are positive and may change from line to line.

Symbol	Meaning	Scope / convention
$\rho=\beta+i\gamma$	Nontrivial zero of ζ	Global
Θ	$\sup\{\Re\rho:\zeta(\rho)=0\}$	Global; RH means $\Theta=1/2$
γ_1	Lowest positive ordinate of a zero of ζ	$14.134725\ldots$
λ_*	e^{π/γ_1}	Sharp threshold of the counting mechanism
θ_*	$\lambda_*/2$	Prefix-dependency cutoff
M, M_0	Mertens function; $M_0=M+2$ in the KP normalization	Global
$I_\nu=(x_{0,\nu},x_{1,\nu}]$	Completed excursion; also $(u_\nu,v_\nu]$	Excursion-indexed
σ_ν	Orientation of I_ν, in $\{\pm1\}$	Reserved for orientation only

Symbol	Meaning	Scope / convention
ℓ_ν	$x_{1,\nu} - x_{0,\nu}$	Excursion length
H_ν, τ_ν	Height and first apex location	Excursion geometry
a_ν, b_ν	$\tau_\nu - x_{0,\nu}$ and $x_{1,\nu} - \tau_\nu$	Excursion geometry
$\mathcal{Z} = [p, q]$	Zero run between consecutive excursions	Never denoted by R
$u(x), L_x$	Last zero $\le x$ and terminal-prefix length	Terminal block
$M_{\mathrm{s}}, M_{\mathrm{sk}}$	Regularized and unregularized skeletons	Global curves
$\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$	Positive, signed-negative, and net area curves	Global curves
$\widetilde{A}$	$\int_0^x M(t)\,dt$	Mellin section; $\widetilde{A}(1) = 0$
S_r	Oriented partial sum inside a specified block	Local; orientation is σ_ν
$C_h, C_h^{(\nu)}$	Lag correlations	Block / excursion context
q_ν, Q_ν	Number of active steps; quadratic mass	Excursion data
β_ν	$\sum_{n \in I_\nu} \lvert M(n)\rvert$	Excursion absolute mass
$\mathcal{E}$	Chord-energy functional	Global
$\mathcal{F}, E_{\mathcal{F}}(X)$	Excursion family and its counting function	Counting sections
$\mathfrak{h}, \mathcal{F}_{\mathfrak{h}}$	Horizon and crossing family	Horizon theorem
$N_\eta(Y)$	KP large-excursion count	Fixed η
μ_u	$\mu \mathbf{1}_{[1,u]}$	Prefix formulas
$\Delta_k(u,t)$	$M(t/k) - M(u/k)$	Channel descent
$\mathcal{R}_K(u,t)$	Deep remainder below channel K	Channel descent; $2 \le K \le u$ when descended
α	Exponent in $\lvert M(q)\rvert/q^\alpha$	Channel descent only
$\mathcal{S}(x), L^{\#}(I), r_I(t)$	Squarefree clock, compressed length, support time	Catalan comparison
p_{Cat}	Primitive fixed-mask Catalan upper-tail weight	Reference law, not a probability model for M"obius
U	Smooth/large-prime level	Not an excursion start
$M_U^{\mathrm{sm}}, M_U^{\mathrm{lp}}$	U^2-smooth and large-prime parts	Large-prime section
$R = t/U^2$	Dimensionless large-prime ratio	Only in that section
$\widetilde{m}(y)$	$\sum_{a \le y} \mu(a)/a$	Large-prime section
$s = \varsigma + i\tau$	Complex analytic variable	$\varsigma = \Re s$, $\tau = \Im s$
$e(x)$	$e^{2\pi i x}$	Fourier convention
$D_I(s)$	Dirichlet block of a completed excursion	Mellin / block-series section
J	A leg/chord when locally declared	Local only
R	Lag cutoff, ratio, or other local scalar when declared	Never a zero run
j, k	Integer indices/depths	Fixed whenever a theorem says "fixed"
λ, ε	Auxiliary horizon / loss parameters	Fixed before the relevant limit

3. Excursions and the regularized skeleton

Let μ be the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad x \ge 0, \tag{3.1}$$

piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$ at the integers. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [6] computed $M(x)$ for all $x \le 10^{16}$. The area and quadratic functionals of lattice excursions are classical [20, 16, 19, 8, 2, 1]; here they are applied to the deterministic Möbius walk.

Definition 3.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}]$$

with integer endpoints, $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \geq 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run* $[x_{1,\nu}, x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. The first excursion is $(0, 2]$. Since M assumes positive and negative values at arbitrarily large arguments (Remark 7.6) and its integer jumps belong to $\{-1, 0, 1\}$, between sufficiently ordered values of opposite sign it must vanish at an integer. Hence M has infinitely many zeros and every excursion is finite. We also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu} < n < x_{1,\nu}} |M(n)|,$$
$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu.$$

Every excursion has $H_\nu \geq 1$ and $\ell_\nu \geq 2$.

Definition 3.2 (Completed excursions and open prefixes)**.** An excursion in the sense of Definition 3.1 is always *completed*: both $M(x_{0,\nu}) = 0$ and $M(x_{1,\nu}) = 0$ are part of the definition. A segment $(u, x]$ with $M(u) = 0$ and $M(n) \neq 0$ for $u < n \leq x$, with no return condition at x, is an *open prefix*; the terminal block of Definition 3.6 is one. Any estimate whose proof uses the backward bound $|M(u+r)| \leq x_{1,\nu} - u - r$ holds for completed excursions only and is never applied to an open prefix.

Lemma 3.3 (Height and step count)**.** *For every excursion,*

$$H_\nu \leq \min(a_\nu, b_\nu) \leq \frac{\ell_\nu}{2}. \tag{3.2}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 3.4 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 3.5 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \leq t \leq \tau_\nu, \\[2ex] \sigma_\nu \sqrt{H_\nu}\, \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \leq t \leq x_{1,\nu}, \end{cases} \tag{3.3}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. Here "regularized" refers only to the normalization of the apex from H_ν to $H_\nu^{1/2}$; no analytic smoothing operation is intended. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 3.6 (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since M has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u < n \leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}_{\mathrm{term}}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big].$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real x with last zero $u(x) \le \lfloor x \rfloor$ define

$$\beta_{\mathrm{term}}(x) := \sum_{u(x)<n\le\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\le\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x\rfloor - u(x).$$

These depend only on M.

Definition 3.7 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\,dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\,dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\,dt. \tag{3.4}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign. The curve $\mathcal{A}_-$ records negative area with its sign; it is not the absolute value of the negative area.

Remark 3.8 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0,\infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \ge 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 5 concern the Möbius increments and do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization.

4. Exact formulas

Lemma 4.1 (Increment per excursion)**.** *For every complete excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu \sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{4.1}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. For the discrete sum, the ascending values at $j = 1,\dots,a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\le a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$; the descending values at $\tau_\nu + r$, $r = 1,\dots,b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\le b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu-1)/2$. □

Proposition 4.2 (Explicit form of the three curves)**.** *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x-x_{0,\nu})^2}{2a_\nu}, & x \le \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & x \ge \tau_\nu. \end{cases} \tag{4.2}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\le k} \sigma_\nu \frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\le k\\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\le k\\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{4.3}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (3.3); the endpoint formulas follow from Lemma 4.1. □

A point y is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of y and strictly concave on the other, within distance δ. The second derivative need not exist at y.

Lemma 4.3 (Critical geometry of $\mathcal{A}$)**.** *Let $\mathcal{Z} = [p,q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then $\mathcal{Z}$ is an extremal plateau; it is a strict local extremum exactly when $p = q$.*

(iii) *If $\sigma_+ = \sigma_-$, then $\mathcal{A}$ has no local extremum on $\mathcal{Z}$. Its concavity has sign $-\sigma_-$ immediately to the left and $+\sigma_+$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of R is a strict inflection, and the change of concavity is carried by the whole plateau.*

(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. On R, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

5. Internal correlations of an excursion

For a real block $x_1, \dots, x_L$ put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{5.1}$$

Lemma 5.1 (Triangular identity)**.** *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{5.2}$$

Proof. Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j (L + 1 - \max(i, j)).$$

Write $\max(i,j) = \frac{1}{2}(i + j + |i - j|)$. The part containing $|i - j|$ is $\frac{1}{2}\sum_{i,j} x_i x_j |i - j| = \sum_{h=1}^{L-1} hC_h$. The remaining part is $(L + 1)S_L^2 - S_L \sum_i i x_i$, and $\sum_i i x_i = LS_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h hC_h$; subtract S_L^2. □

Corollary 5.2 (Bridge case)**.** *If $S_L = 0$, then*

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h. \tag{5.3}$$

Remark 5.3 (Hermitian form). For complex x_j and $C_h = \sum_{j=1}^{L-h} x_j \overline{x_{j+h}}$, the same computation gives

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L} \sum_{t=1}^{L-1} S_t \right).$$

For a complete excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{5.4}$$

Then $S_t > 0$ for $1 \le t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} hC_h^{(\nu)}. \tag{5.5}$$

Lemma 5.4 (Zeroth moment and extreme lag)**.** *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu - 1}^{(\nu)} = -1, \tag{5.6}$$

and q_ν is even and at least 2.

Proof. $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 5.5 (Quadratic decomposition by excursions)**.** *For real $x \geq 1$ let $u(x)$ and L_x be as in Definition 3.6, and for $1 \leq j \leq L_x$ put $y_j = \mu(u(x)+j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x-h} y_j y_{j+h}$. Then*

$$\sum_{n\leq x} M(n)^2 = \sum_{\nu: x_{1,\nu}\leq u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{5.7}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x\rfloor)^2 + M(\lfloor x\rfloor)\sum_{t=1}^{L_x-1} M(u(x)+t) - \sum_{h=1}^{L_x-1} hC_h^{\mathrm{term}}, \tag{5.8}$$

with $Q_{\mathrm{term}}(x) = 0$ if $L_x = 0$. At a zero x of M,

$$\sum_{n\leq x} M(n)^2 = \sum_{x_{1,\nu}\leq x} Q_\nu = -\sum_{x_{1,\nu}\leq x}\sum_h hC_h^{(\nu)}. \tag{5.9}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 5.1 to the terminal block and add its final term $S_{L_x}^2$. □

Lemma 5.6 (Elementary bound for an open terminal block)**.** *Let u be a zero of M and $L \geq 0$. Then $|M(u+r)| \leq r$ for $1 \leq r \leq L$, and*

$$\sum_{r=1}^{L} M(u+r)^2 \leq \frac{L(L+1)(2L+1)}{6}. \tag{5.10}$$

In particular $Q_{\mathrm{term}}(x) \ll L_x^3$. The bridge bound (5.17)*, whose leading coefficient is $1/12$ rather than $1/3$, applies only to complete excursions.*

Proof. $M(u+r) = \sum_{j\leq r}\mu(u+j)$ with $|\mu| \leq 1$; square and sum. □

Thus the quadratic sum sees only lag correlations internal to excursions: pairs of Möbius increments in different completed excursions do not contribute.

Let

$$\beta_\nu = \sum_{t=1}^{\ell_\nu-1} S_t = \sum_{n\in I_\nu} |M(n)|. \tag{5.11}$$

Cauchy–Schwarz gives

$$\beta_\nu^2 \leq (\ell_\nu - 1)Q_\nu, \tag{5.12}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu-1} h(2D_h^{(\nu)} - P_h^{(\nu)}). \tag{5.13}$$

Proposition 5.7 (Sharp bounds at fixed length and height)**.** *Let $S_0 = S_L = 0$, $S_t \geq 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ and $H = \max S_t$. Then $L \geq 2H$ and*

$$H^2 + L - 2H \leq \beta \leq H(L-H), \tag{5.14}$$

$$\frac{2H^3+H}{3} + L - 2H \leq Q \leq \frac{2H^3+H}{3} + (L-2H)H^2. \tag{5.15}$$

All four bounds are attained. In particular

$$Q \geq \frac{2H^3 + H}{3}, \qquad Q \geq L - 1. \tag{5.16}$$

Proof. Before and after the first visit to H the path visits each level $1, \dots, H-1$. Together with one visit to H, these $2H-1$ designated interior times contribute $2(1 + \cdots + (H-1)) + H = H^2$ to β and $2\sum_{j<H} j^2 + H^2 = (2H^3 + H)/3$ to Q. The remaining $L - 2H$ interior times lie at heights between 1 and H; placing them all at height 1, or all at height H, attains the bounds. □

Corollary 5.8 (Sharp fixed-length bounds)**.** *For $L \geq 2$,*

$$\beta \leq \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \leq \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases}$$

and in particular

$$Q \leq \frac{L^3 + 2L}{12}. \tag{5.17}$$

Both bounds are attained by the tent path, with one horizontal step at maximal height when L is odd.

Proof. Use $S_t \leq \min(t, L-t)$ and sum this bound and its square. □

6. Where ζ enters

The black curve depends nonlinearly on M. The dependence enters through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$; no closed expression of its Mellin transform in terms of ζ is known. The nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of M. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer x,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\geq 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{6.1}$$

[21, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has $\widetilde{A}(1) = 0$ (the value of M at the single point 1 does not affect the integral). Integration by parts therefore gives, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{6.2}$$

Shifting the contour formally, the residues are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{6.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s = 0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s = -1$. The factor $(\rho+1)^{-1}$ improves convergence, but RH and simplicity alone do not control $1/\zeta'(\rho)$ well enough to turn (6.3) into an identity with an untruncated sum; results of this type require additional input of Gonek–Hejhal type [15]. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms.

Proposition 6.1 (Dipole identity)**.** *For each excursion let its* legs *be the integer sets $J = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ and $(\tau_\nu, v_\nu] \cap \mathbb{Z}$, with midpoints $c_J = (u_\nu + \tau_\nu)/2$ and $(\tau_\nu + v_\nu)/2$. Then for every zero v_k,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{J\subset(0,v_k]} \sum_{m\in J} \mu(m)\,(m - c_J). \tag{6.4}$$

Proof. Since M is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma_\nu H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma_\nu H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_J)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma_\nu H b/2$ and $\int_\tau^v M = b\,\sigma_\nu H + \sum_{m=\tau+1}^{v} \mu(m)(v-m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma_\nu H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_J$. On zero runs both integrands vanish. Sum over legs. □

7. Bounds for the area curves

Lemma 7.1 (Transfer)**.** *Let $x \geq 1$ and let $x^* \geq x$ be a zero of M (in a computation truncated at $N \geq x$, take $x^* = N$ and the terminal skeleton of Definition 3.6). If $|M(n)| \leq C_0\, n^{1/2}$ for all $1 \leq n \leq x^*$ (so necessarily $C_0 \geq 1$), then*

$$|M_{\mathrm{s}}(t)| \leq C_0^{1/2} t^{1/4} \quad (0 \leq t \leq x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq \tfrac{4}{5}\, C_0^{1/2} x^{5/4}. \tag{7.1}$$

Proof. For $t \geq \tau_\nu$, $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq C_0^{1/2} \tau_\nu^{1/4} \leq C_0^{1/2} t^{1/4}$. For $u_\nu \leq t \leq \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $C_0^{1/2} t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $C_0^{1/2} u_\nu^{1/4} \geq 0$ and $C_0^{1/2} \tau_\nu^{1/4} \geq \sqrt{H_\nu}$ (as $\tau_\nu \leq x^*$) are at least as high; that chord lies below $C_0^{1/2} t^{1/4}$ by concavity. Integrate. □

Lemma 7.2 (A concave majorant)**.** *Let $\phi(y) = (\log y)^{3/5} (\log\log y)^{-1/5}$ and, for $\kappa > 0$, $F(y) = y^{1/2} e^{-\kappa\phi(y)}$. There is $y_0 = y_0(\kappa)$ such that F is increasing and concave on $[y_0, \infty)$.*

Proof. With $y = e^s$ and $g(s) = s^{3/5}(\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \frac{1}{2} - \kappa g'(s)$. Then

$$y F'(y) = F\, h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big).$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \frac{1}{2}$ and $h^2 - h - \kappa g'' \to -\frac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large y. □

Corollary 7.3. *From the Vinogradov–Korobov bound* [22]

$$M(x) \ll x \exp\big(-c\,\phi(x)\big) \tag{7.2}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\big(-\tfrac{c}{2}\,\phi(x)\big). \tag{7.3}$$

Proof. Let K be such that $|M(y)| \leq K y\, e^{-c\phi(y)}$ for $y \geq y_1$, and let F be as in Lemma 7.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \leq K^{1/2} F(y)$ for $y \geq y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \geq Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \leq K^{1/2} F(\tau_\nu)$; the bound is taken at the apex τ_ν, which lies in the excursion, and not at its future endpoint v_ν. For $t \geq \tau_\nu$ monotonicity of F gives $|M_{\mathrm{s}}(t)| \leq \sqrt{H_\nu} \leq K^{1/2} F(t)$. For $u_\nu \leq t \leq \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2}F(u_\nu))$ to $(\tau_\nu, K^{1/2}F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2}F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \leq K^{1/2} F(t)$ for all $t \geq Y_0$ outside the finitely many early excursions, and since F is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \leq O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \leq O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \quad \square$$

Corollary 7.4. *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{7.4}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false.

Remark 7.5 (Endpoint-indexed bounds). From $\ell_\nu \geq 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces only $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$. This bound is indexed by the *future* endpoint v_ν of the excursion; it gives no bound for $M(x)$ in terms of x at interior points, and hence no zero-free half-plane. Whether (7.4) is equivalent to RH is open.

Remark 7.6 (Oscillation of the heights). The Mertens conjecture is false [17, 10]; the current bounds are due to Hurst [6], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point n with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by v_ν does not follow, since v_ν may be much larger than n.

8. Global criteria equivalent to RH

The Mellin identity and the global L^1/L^2 equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas repackage them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x)x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{8.1}$$

If $\int_1^\infty |M(x)|x^{-\varsigma-1}dx < \infty$ for every $\varsigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so ζ has no zeros there.

Proposition 8.1 (L^1 criterion). *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \leq x} \beta_\nu + \beta_{\mathrm{term}}(x) = \sum_{n\leq x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{8.2}$$

Proof. RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Conversely, for $\varsigma > 1/2$ choose $0 < \varepsilon < \varsigma - 1/2$; partial summation gives $\sum_n |M(n)|n^{-\varsigma-1} < \infty$, and since M is constant on $[n, n+1)$ the remark after (8.1) applies. □

Lemma 8.2 (Pruning). *$\beta_\nu \leq H_\nu(\ell_\nu - 1) \leq \ell_\nu^2/2$, hence*

$$\sum_{v_\nu \leq x,\ \ell_\nu \leq L_0} \beta_\nu \leq \frac{xL_0}{2}. \tag{8.3}$$

Corollary 8.3 (Reduction to long excursions). *For fixed $\delta \geq 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \leq x \\ \ell_\nu > L_0}} \beta_\nu + \mathbf{1}_{\{L_x > L_0\}}\beta_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \textit{for every } \varepsilon > 0. \tag{8.4}$$

At most x/L_0 long excursions are involved.

Proof. If RH holds, the left side is at most $\sum_{n\leq x}|M(n)| \ll x^{3/2+\varepsilon}$. Conversely, the complete excursions with $\ell_\nu \leq L_0$ contribute at most $xL_0/2 \leq x^{3/2}$ by Lemma 8.2, and if $L_x \leq L_0$ the terminal block contributes $\beta_{\mathrm{term}}(x) \leq \sum_{r\leq L_x} r = L_x(L_x+1)/2 \leq L_0^2 \leq x$. Hence $\sum_{n\leq x}|M(n)| \ll x^{3/2+\varepsilon}$, and Proposition 8.1 applies. □

Remark 8.4 (The $\sqrt{x}$ frontier). The cut may depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\leq x^{3/2+\varepsilon/2}/2$. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\beta_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

Heuristic 8.5 (Renewal tail)**.** If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x : \ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{8.5}$$

and with $\beta_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (8.2), so the terminal scales $L \asymp x$ dominate the L^1 sum.

Theorem 8.6 (Correlation criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n\le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x} \sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{8.6}$$

Proof. Under RH, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n\le x} |M(n)| \le x^{1/2}\Big(\sum_{n\le x} M(n)^2\Big)^{1/2} \ll x^{3/2+\varepsilon/2};$$

apply Proposition 8.1. □

Corollary 8.7 (Local sufficient conditions)**.** *Suppose that for every $\varepsilon > 0$ there is C_ε such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \text{for every complete excursion with } \ell_\nu > v_\nu^{1/2}, \tag{8.7}$$

Then RH holds.

Proof. Let $\varepsilon_0 \in (0,1)$, put $\varepsilon = \varepsilon_0/4$, and let C_ε be as in the hypothesis. Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (5.17) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon (\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a complete excursion, and (5.17) does not apply to it. By Lemma 5.6, $Q_{\mathrm{term}}(x) \le L_x(L_x+1)(2L_x+1)/6$, which is $O(x^{3/2})$ if $L_x \le x^{1/2}$. If $L_x > x^{1/2}$, then $M(\lfloor x \rfloor) \ne 0$. Let $I_\nu = (u(x), v_\nu]$ be the unique completed excursion whose open prefix contains $\lfloor x \rfloor$; since all summands are nonnegative, $Q_{\mathrm{term}}(x) \le Q_\nu$. Fix $A = 1 + \varepsilon$. By Corollary 9.8, $v_\nu \le u(x)^A \le x^A$ once x is large, so $\ell_\nu \le x^A$. If $\ell_\nu > v_\nu^{1/2}$ the hypothesis gives $Q_\nu \le C_\varepsilon x^{A(2+\varepsilon)}$; otherwise (5.17) gives $Q_\nu \ll \ell_\nu^3 \le v_\nu^{3/2} \le x^{3A/2}$. In all cases $Q_{\mathrm{term}}(x) \ll_\varepsilon x^{(1+\varepsilon)(2+\varepsilon)} \le x^{2+4\varepsilon} = x^{2+\varepsilon_0}$. This step uses the arithmetic input of Corollary 9.8: for a general family of disjoint intervals, the future return of the terminal block is not controlled by x, and a bound for complete excursions would not control $Q_{\mathrm{term}}(x)$. Altogether $\sum_{n\le x} M(n)^2 \ll_{\varepsilon_0} x^{2+\varepsilon_0}$ for every $\varepsilon_0 \in (0,1)$, and Theorem 8.6 gives RH. □

Heuristic 8.8 (Ballistic excursions)**.** In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded k, then by (5.16) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version of (8.7) fails for every $\varepsilon \in (0,1)$. Such excursions have $\ell_\nu \ll v_\nu^{1/2}$ and do not affect the pruned condition (8.7); this is why the pruning in Corollary 8.7 is needed.

Theorem 8.9 (Energy criterion)**.** *With u the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x} |M(n)|$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u) H_{\mathrm{run}}(x)\Big). \tag{8.8}$$

For a complete excursion let J be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By (4.1), $J' = \sigma_\nu \sqrt{H_\nu}/2$, so $\frac{1}{4} H_\nu \ell_\nu = \int_{I_\nu} (J')^2$ is the Dirichlet energy of the chord. Then

$$\sum_{n\le x} |M(n)| \le 4\mathcal{E}(x) \le x \max_{n\le x} |M(n)|, \tag{8.9}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\beta_\nu \le H_\nu \ell_\nu$, $\beta_{\mathrm{term}}(x) \le (x-u)H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 8.1. □

8.1. **Exact logarithmic exponents from Pintz's mean and late-maximal-order theorem.** The criteria above are upper-bound equivalences. Theorems 2.1–2.2 of Pintz [18] determine, in logarithmic scale, the exponent of the mean modulus and of the late maximal order of $|M|$ in terms of the rightmost zeros of ζ. We record the consequence for the excursion quantities used in this paper.

Theorem 8.10 (Pintz, logarithmic mean and late maximal order [18, Theorems 2.1–2.2])**.** *Let* $\Theta = \sup\{\Re\rho : \zeta(\rho) = 0\}$. *Then*

$$D_M(X) := \frac{1}{X}\int_1^X |M(t)|\,dt = X^{\Theta+o(1)}.$$

Moreover, for every fixed $\delta > 0$,

$$S_{M,\delta}(X) := \max_{X^{1-\delta}\le t\le X} |M(t)| = X^{\Theta+o(1)}.$$

Here the equalities are in logarithmic scale, equivalently

$$\frac{\log D_M(X)}{\log X} \to \Theta, \qquad \frac{\log S_{M,\delta}(X)}{\log X} \to \Theta.$$

Corollary 8.11 (Exact exponents of excursion area and quadratic mass)**.** *Put*

$$B_1(X) := \sum_{n\le X} |M(n)|, \qquad B_2(X) := \sum_{n\le X} M(n)^2.$$

Then

$$B_1(X) = X^{1+\Theta+o(1)}, \qquad B_2(X) = X^{1+2\Theta+o(1)}, \qquad \mathcal{E}(X) = X^{1+\Theta+o(1)}. \tag{8.10}$$

Consequently

$$\Theta = \lim_{X\to\infty}\left(\frac{\log B_1(X)}{\log X} - 1\right) = \frac{1}{2}\lim_{X\to\infty}\left(\frac{\log B_2(X)}{\log X} - 1\right) = \lim_{X\to\infty}\left(\frac{\log \mathcal{E}(X)}{\log X} - 1\right). \tag{8.11}$$

In particular RH is equivalent to any one of the three logarithmic exponents being respectively $3/2$, 2, *and* $3/2$.

Proof. Let $N = \lfloor X \rfloor$. Since $M(t) = M(n)$ on $[n, n+1)$,

$$\int_1^X |M(t)|\,dt = \sum_{n=1}^{N-1} |M(n)| + (X-N)|M(N)|,$$

so

$$B_1(X) = \int_1^X |M(t)|\,dt + O(|M(N)| + 1).$$

The boundary term is $O(X)$, whereas Theorem 8.10 gives $\int_1^X |M(t)|\,dt = X^{1+\Theta+o(1)}$ and $\Theta \ge 1/2$; hence $B_1(X) = X^{1+\Theta+o(1)}$. We first note that

$$\max_{n\le X} |M(n)| = X^{\Theta+o(1)}. \tag{8.12}$$

The lower bound follows from $S_{M,\delta}$. For the upper bound, if $\Theta < 1$, fix once and for all $\delta \in (1-\Theta, 1)$ and split at $X^{1-\delta}$: the early range is trivially at most $X^{1-\delta}$ and the late range is controlled by Theorem 8.10; when $\Theta = 1$ the trivial bound $|M(n)| \le n$ suffices.

Cauchy–Schwarz and (8.12) now give

$$\frac{B_1(X)^2}{X} \le B_2(X) \le X \max_{n\le X} |M(n)|^2,$$

so both sides have exponent $1+2\Theta$. Finally (8.9) squeezes $4\mathcal{E}(X)$ between $B_1(X)$ and

$$X \max_{n\le X} |M(n)|,$$

both of exponent $1+\Theta$. □

Corollary 8.12 (Length localization of the area and quadratic mass)**.** *Fix $0 \le \alpha < \Theta$. Among the complete excursions ending at most X, the total contribution of those with $\ell_\nu \le X^\alpha$ is negligible:*

$$\sum_{\substack{v_\nu \le X \\ \ell_\nu \le X^\alpha}} \beta_\nu + \mathbf{1}_{\{L_X \le X^\alpha\}} \beta_{\mathrm{term}}(X) = o(B_1(X)),$$

and

$$\sum_{\substack{v_\nu \le X \\ \ell_\nu \le X^\alpha}} Q_\nu + \mathbf{1}_{\{L_X \le X^\alpha\}} Q_{\mathrm{term}}(X) = o(B_2(X)).$$

Thus, on the logarithmic scale forced by the zeta zeros, essentially all L^1 and L^2 mass is carried by excursion blocks longer than X^α for every $\alpha < \Theta$.

Proof. Lemma 8.2 bounds the first complete-excursion sum by $O(X^{1+\alpha})$, while the short terminal block contributes $O(X^{2\alpha})$. Both are $o(X^{1+\Theta+o(1)})$ when $\alpha < \Theta$. For the quadratic mass use (5.17) and disjointness:

$$\sum_{\ell_\nu \le X^\alpha} Q_\nu \ll X^{1+2\alpha}, \qquad Q_{\mathrm{term}}(X) \ll X^{3\alpha} \quad (L_X \le X^\alpha).$$

Because $\alpha < \Theta \le 1$, both exponents are strictly below $1+2\Theta$. □

9. Lacunary excursion families and horizon-localized criteria

9.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 9.6). For a family $\mathcal{F}$ of excursions and $X \ge 2$ put

$$E_{\mathcal{F}}(X) := \#\{\nu \in \mathcal{F} : x_{0,\nu} \le X\}.$$

For a function $\mathfrak{h} : [0,\infty) \to [0,\infty]$, the family of excursions *crossing the horizon h* is

$$\mathcal{F}_{\mathfrak{h}} := \{\nu : x_{1,\nu} > \mathfrak{h}(x_{0,\nu})\}.$$

For $A > 1$ the excursion I_ν is *A-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$, i.e. $\nu \in \mathcal{F}_{x^A}$, and *$A$-ordinary* otherwise; for $A = 2$ we say *square-exceptional* and *square-ordinary*. For $\lambda > 1$ it is *λ-long* if $x_{1,\nu} > \lambda x_{0,\nu}$. We write $E_A = E_{\mathcal{F}_{x^A}}$, $E_\square = E_2$ and $E_\lambda = E_{\mathcal{F}_{\lambda x}}$; the context separates E_A (A an exponent) from E_λ (λ a factor).

Lemma 9.1 (Power lacunarity)**.** *For every $A > 1$ and $X \ge 2$,*

$$E_A(X) \le 2 + \frac{1}{\log A} \log \frac{\log X}{\log 2}. \tag{9.1}$$

In particular

$$E_\square(X) \le 2 + \log_2 \frac{\log X}{\log 2}, \qquad E_\square(X) \le 4 \log\log X \quad (X \ge e^e). \tag{9.2}$$

Proof. No excursion starts at 1, since $M(1) = 1$. List the A-exceptional starts ≥ 2 as $2 \leq a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \geq x_{1,j} > a_j^A$ and $\log a_j \geq A^{j-1} \log 2$. If $a_j \leq X$ then $j \leq 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0, 2]$. For the second bound in (9.2), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \leq 1.443 \log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

Lemma 9.2 (Fractional square horizon)**.** *For $0 < c \leq 1$ and $X \geq 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > cx_{0,\nu}^2\} \leq \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{9.3}$$

Consequently, the range $cx_0^2 < t \leq x_0^2$ of the excursions starting at most X is visited by at most $O_c(\log\log X)$ excursions.

Proof. At most $\lceil 2/c \rceil \leq 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \geq 2$. Then $a_{j+1} \geq x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2,$$

and $\log b_j \geq 2^{j-1} \log 2$. Since $b_j \leq cX \leq X$, the number of such j is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \geq t > cx_0^2$. □

Lemma 9.3 (Superlinear horizons)**.** *Let $\psi : [1, \infty) \to (0, \infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X).$$

Proof. Given $K > 0$ choose x_K with $\psi(x) \geq e^K$ for $x \geq x_K$. At most $x_K + 1$ members start below x_K. For consecutive members $a_j < a_{j+1}$ starting at or above x_K, $a_{j+1} \geq x_{1,j} > a_j\psi(a_j) \geq e^K a_j$, so $\log a_j$ increases by more than K at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \leq 1/K$ for every K. □

Lemma 9.4 (Linear lacunarity)**.** *For every $\lambda > 1$ and $X \geq 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \leq 2 + \frac{\log(X/2)}{\log \lambda}. \tag{9.4}$$

Proof. For starts $a_j \geq 2$, $a_{j+1} \geq x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \geq 2$; add 1 for $(0, 2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

9.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, its translation from $M_0 = M + 2$ to M, and the consequences for excursions.

Theorem 9.5 (Kaczorowski–Pintz [9])**.** *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$, the function M_a of [9] at $a = 0$. Let $V(f, Y)$ be defined as in [9], through the largest number of points of $[1, Y]$ at which f is nonzero with alternating signs; the number of sign changes of f on $[1, Y]$ is at least $V(f, Y) - 1$, which does not affect the asymptotic statements below. Then:*

(i) [9, Corollary 4, p. 178, (2.13)–(2.15)]

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \tag{9.5}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every fixed $\eta > 0$: there exist ordered points $t_1 < \cdots < t_N \leq Y$, with $N \geq (\gamma_1/\pi - o(1)) \log Y$, whose signs alternate and which satisfy $|M_0(t_j)| \geq t_j^{\Theta-\eta}$;

(ii) *The local statement comes from Theorem 1, (2.2)–(2.3), and Corollary 1 of* [9]*; in particular, for every* $\delta > 0$ *and* $\eta > 0$ *there is* $Y_0(\delta, \eta)$ *such that for* $Y > Y_0(\delta, \eta)$ *the interval* $[Y^{1-\delta}, Y]$ *contains two points* t_+, t_- *with* $M_0(t_+) \geq t_+^{\Theta-\eta}$ *and* $M_0(t_-) \leq -t_-^{\Theta-\eta}$*. (In* [9] *both losses are governed by a single parameter* ε*; the form above follows by taking* $\varepsilon = \min\{\delta, \eta\}$*. Indeed* $Y^{1-\varepsilon} \geq Y^{1-\delta}$ *and* $x^{\Theta-\varepsilon} \geq x^{\Theta-\eta}$*, so the original interval and magnitude bound are contained in the formulation above.)*

Lemma 9.6 (From M_0 to M)**.** *If* $|M_0(t)| > 2$*, then* $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ *and* $|M(t)| \geq |M_0(t)| - 2$*. In particular, for points with* $|M_0(t)| \geq t^{\Theta-\eta}$ *and* t *large,* $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ *and* $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$*.*

Proof. $M_0 > 2$ gives $M > 0$; $M_0 < -2$ gives $M < -4$. The second assertion uses $t^{\Theta-\eta} \to \infty$ for $\eta < \Theta$. □

Corollary 9.7 (Distinct large excursions)**.** *If* $\Theta > 1/2$ *and* $0 < \eta < \Theta - 1/2$*, then*

$$N_\eta(Y) \geq \left(\frac{\gamma_1}{\pi} - o(1) \right) \log Y, \tag{9.6}$$

where $N_\eta(Y)$ *counts distinct excursions starting at most* Y *that contain a point* $t \leq Y$ *with* $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$*.*

Proof. Take the alternating points of Theorem 9.5(i) in $[1, Y]$, discarding the boundedly many below the threshold of Lemma 9.6. By that lemma, consecutive points $t < t'$ have $M(t)$ and $M(t')$ of opposite signs. Since M is integer valued with steps $0, \pm 1$, it vanishes at some integer in (t, t'), so t and t' lie in different excursions. Each point with $M(t) \neq 0$ belongs to exactly one excursion, and excursions are ordered, so the excursions containing the ordered points are pairwise distinct. Their starts are below the points, hence at most Y, and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$ at each point. □

Corollary 9.8 (Multiplicative localization of Mertens returns)**.** *For every fixed* $A > 1$ *there exists* U_A *such that every excursion starting at* $u \geq U_A$ *ends at* $v \leq u^A$*. Equivalently,* M *has only finitely many* A*-exceptional excursions; in particular only finitely many square-exceptional ones.*

Proof. Put $\delta = 1 - 1/A$ and $Y = u^A$, so that $[Y^{1-\delta}, Y] = [u, u^A]$. Fix once and for all $\eta = \Theta/2$. After this choice the resulting threshold depends only on A; denote it by U_A. For $u \geq U_A$, Theorem 9.5(ii) and Lemma 9.6 give $t_+, t_- \in [u, u^A]$ with $M(t_+) > 0 > M(t_-)$. Neither point lies in $[u, u+1)$, because $M(u) = 0$; so $u < \lfloor t_\pm \rfloor$. Since M changes by $0, \pm 1$ at each integer, it vanishes at some integer n strictly between $\lfloor t_+ \rfloor$ and $\lfloor t_- \rfloor$, and $u < n < u^A$. If $u = x_{0,\nu}$, the first return satisfies $x_{1,\nu} \leq n < u^A$. □

Remark 9.9. A bare sign change of $M_0 = M + 2$ would not force a zero of M. The decisive point is that the local sign change is combined with oscillation of size $t^{\Theta-\eta} \to \infty$, so that Lemma 9.6 applies to both witnesses.

Remark 9.10 (Geometric counts and arithmetic localization). For the actual Mertens excursions Corollary 9.8 is stronger than Lemma 9.1: each A-exceptional family is finite. The geometric lemmas remain separate results because they are elementary, independent of ζ, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 9.6 use: the former needs only a count below $(\gamma_1/\pi) \log X$, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

9.3. **Horizon localization.**

Theorem 9.11 (Horizon-localized RH criterion)**.** *Let* $\mathfrak{h} : [0, \infty) \to [0, \infty]$ *be a function, let* $\mathcal{F}$ *be a family of excursions, and suppose that for some* $\kappa < \gamma_1/\pi$

$$E_{\mathcal{F} \cup \mathcal{F}_{\mathfrak{h}}}(X) \leq (\kappa + o(1)) \log X \qquad (X \to \infty). \tag{9.7}$$

Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t \in I_\nu,\ \nu \notin \mathcal{F},\ t \le \mathfrak{h}(x_{0,\nu}). \tag{9.8}$$

No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $\mathfrak{h}(x_0)$.

Proof. RH implies (9.8) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By (9.6) there are at least $(\gamma_1/\pi - o(1)) \log X$ distinct excursions starting at most X, each containing a witness t with $|M(t)| \ge \frac{1}{2} t^{\Theta-\eta}$. Consider such an excursion ν with witness t. If $\nu \notin \mathcal{F}$ and $t \le \mathfrak{h}(x_{0,\nu})$, the witness lies in the range of (9.8). Otherwise $\nu \in \mathcal{F}$, or $t > \mathfrak{h}(x_{0,\nu})$, in which case $x_{1,\nu} > t > \mathfrak{h}(x_{0,\nu})$ (as $M(t) \ne 0$) and $\nu \in \mathcal{F}_{\mathfrak{h}}$. By (9.7) at least

$$\Big(\frac{\gamma_1}{\pi} - \kappa - o(1)\Big) \log X \longrightarrow \infty$$

witnesses lie in the range of (9.8). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon} \to \infty$, contradicting (9.8). □

The case $\mathfrak{h}(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

Corollary 9.12 (Square horizon)**.** *For every fixed $A > 1$, RH is equivalent to* (9.8) *restricted to A-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2,} \tag{9.9}$$

A separate application of Theorem 9.11, with no excursions discarded and horizon $\mathfrak{h}(x) = x^2$, gives the second RH-equivalent statement

$$|M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad (x_{0,\nu} < t \le \min\{x_{1,\nu}, x_{0,\nu}^2\})$$

over all excursions. The two restricted statements are therefore two distinct consequences of the horizon theorem, not an elementary equivalence between the restrictions themselves.

Proof. Take $\mathcal{F} = \mathcal{F}_{x^A}$, $\mathfrak{h} = \infty$, respectively $\mathcal{F} = \emptyset$, $\mathfrak{h}(x) = x^2$, in Theorem 9.11; Lemma 9.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

Corollary 9.13 (Descending horizons)**.** *RH is equivalent to Littlewood's bound $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ on the ranges*

$$x_{0,\nu} < t \le \min\{x_{1,\nu}, \mathfrak{h}(x_{0,\nu})\}$$

over all excursions, for each of the following horizons:

(i) $\mathfrak{h}(x) = cx^2$, *for any fixed $c > 0$;*
(ii) $\mathfrak{h}(x) = x\psi(x)$, *for any ψ with $\psi(x) \to \infty$;*
(iii) $\mathfrak{h}(x) = \lambda x$, *for any fixed $\lambda > \lambda_* := e^{\pi/\gamma_1} = 1.24889\ldots$*

The same holds with "truncate at $\mathfrak{h}(x_0)$" replaced by "discard the excursions with $x_1 > \mathfrak{h}(x_0)$". In particular it is enough to control the excursions with $x_1 \le \frac{5}{4} x_0$.

Proof. Apply Theorem 9.11 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_{\mathfrak{h}}$, $\mathfrak{h} = \infty$). For (i), $c \ge 1$ reduces to Lemma 9.1 and $0 < c < 1$ to Lemma 9.2; $\kappa = 0$. For (ii), Lemma 9.3 gives $\kappa = 0$. For (iii), Lemma 9.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < \gamma_1/\pi$ exactly when $\lambda > \lambda_*$. Finally $5/4 > \lambda_*$. □

At the level of the restricted statements, (i) and (ii) imply (iii) eventually, because their horizons are at least λx_0 for every fixed admissible $\lambda > \lambda_*$ once x_0 is large. The converse does not follow from inclusion of ranges; rather, each of (i)–(iii) is separately RH-equivalent by Theorem 9.11. The finitely many small excursions affect only the constant C_ε. The horizon has been lowered from x_0^2 to λx_0: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

Corollary 9.14 (Two-sided localization of large values)**.** *Suppose $\Theta > 1/2$, and fix $0 < \eta < \Theta - \frac{1}{2}$ and $\lambda > \lambda_*$ before letting $X \to \infty$. Among the excursions starting at $x_0 \le X$, at least*

$$\Big(\frac{\gamma_1}{\pi} - \frac{1}{\log \lambda} - o(1)\Big) \log X$$

contain a point t_* with $|M(t_*)| \ge \frac{1}{2} t_*^{\Theta-\eta}$ and

$$x_0 + \tfrac{1}{2} x_0^{\Theta-\eta} \le t_* \le \lambda x_0.$$

Proof. Of the excursions of Corollary 9.7, those whose witness exceeds λx_0 are λ-long, hence at most $\log X / \log \lambda + 2$ by Lemma 9.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \le t_* - x_0$, so $t_* - x_0 \ge \frac{1}{2} t_*^{\Theta-\eta} \ge \frac{1}{2} x_0^{\Theta-\eta}$. □

Remark 9.15 (Dyadic erosion and the uniform transition). Fix an integer $j \ge 0$ before letting $X \to \infty$. Lemma 9.2 with $c = 2^{-j}$ gives

$$\# \left\{ \nu : x_{0,\nu} \le X,\ x_{1,\nu} > \frac{x_{0,\nu}^2}{2^j} \right\} = O_{2^j}(\log \log X),$$

so Theorem 9.11 successively permits the horizons

$$x_0^2, \quad \frac{x_0^2}{2}, \quad \frac{x_0^2}{4}, \quad \frac{x_0^2}{8}, \dots.$$

The constant in the fixed-c estimate depends on c; hence one may not set $j = j(x_0) \sim \log_2 x_0$ by that lemma alone. If instead $2^{j(x)} = x/\psi(x)$ with $\psi(x) \to \infty$, then the corresponding horizon is $x\psi(x)$ and Lemma 9.3 supplies the uniform $o(\log X)$ crossing count. Thus the rigorous erosion chain is

$$x_0^2 \to \frac{x_0^2}{2} \to \frac{x_0^2}{4} \to \cdots \to x_0 \psi(x_0) \to \lambda x_0, \qquad \lambda > \lambda_*.$$

9.4. **Fixed-mask Catalan compression inside fractional square horizons.** The fractional-square localization has a useful interaction with the exact combinatorics of a completed excursion. The relevant Catalan time is not physical time: zero Möbius increments must first be removed. This support-time distinction is essential below.

Definition 9.16 (Support-time path and fixed-mask Catalan law)**.** Let $I = (x_0, x_1]$ be a complete excursion, and list the nonzero increments in I as

$$x_0 < a_1 < a_2 < \cdots < a_L \le x_1, \qquad \mu(a_j) \ne 0.$$

Put

$$r_I(t) := \#\{j : a_j \le t\}, \qquad x_0 < t \le x_1.$$

For the excursion I_ν, let $\sigma_\nu \in \{\pm 1\}$ be its orientation and define

$$S_0 = 0, \qquad S_r := \sigma_\nu \sum_{j=1}^{r} \mu(a_j) \qquad (1 \le r \le L).$$

Then $L = 2n$ is even and

$$S_0 = S_L = 0, \qquad S_r > 0 \quad (1 \le r < L), \qquad S_r - S_{r-1} \in \{-1, +1\}.$$

Thus $(S_r)_{0\le r\le L}$ is a primitive Dyck path. The *fixed-mask primitive Catalan law* keeps the physical support $\{a_1,\dots,a_L\}$ fixed and assigns to those support positions a uniformly chosen primitive Dyck sign sequence of length L. We denote its probability by $\mathbb{P}_{\mathrm{pCat},I}$.

For the observed excursion one has, exactly,

$$|M(t)| = S_{r_I(t)}, \qquad r_I(t) \le t - x_0 < t. \tag{9.10}$$

The first equality uses the orientation of the excursion and the fact that M is constant across the zero increments.

We first record the global Catalan estimate, since it is useful in its own right and clarifies why a different estimate is needed at the square horizon.

Lemma 9.17 (Catalan maximum tail)**.** *Let $L = 2n$ and let $H_{\max}$ be the maximum of a uniformly chosen Dyck path of length $2n$. For $1 \le h \le n$,*

$$\mathbb{P}_{\mathrm{Cat},L}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le (n+1)\exp\left(-\frac{h^2}{n+h}\right) \le \left(\frac{L}{2}+1\right)e^{-h^2/L}. \tag{9.11}$$

For a uniformly chosen primitive Dyck path of length $L = 2n$,

$$\mathbb{P}_{\mathrm{pCat},L}\{H_{\max} \ge h\} \le \frac{L}{2}\exp\left(-\frac{(h-1)^2}{L-2}\right), \qquad 2 \le h \le n. \tag{9.12}$$

Proof. There are $\mathrm{Cat}_n = (n+1)^{-1}\binom{2n}{n}$ Dyck paths of length $2n$ and $\binom{2n}{n}$ unrestricted bridges. Reflecting a bridge after its first visit to level h gives a bijection between bridges that reach h and unrestricted walks ending at $2h$, hence their number is $\binom{2n}{n+h}$. Therefore

$$\mathbb{P}_{\mathrm{Cat},2n}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}}.$$

Moreover

$$\frac{\binom{2n}{n+h}}{\binom{2n}{n}} = \prod_{j=1}^{h}\frac{n-j+1}{n+j},$$

and $\log(1-x) \le -x$ gives

$$\log\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le -\sum_{j=1}^{h}\frac{2j-1}{n+j} \le -\frac{h^2}{n+h}.$$

Since $n+h \le 2n = L$, (9.11) follows. A primitive path of length $2n$ has the form $+1$, followed by a Dyck path of length $2n-2$, followed by -1; its maximum is one plus the maximum of the interior Dyck path. Applying the preceding bound with $n-1$ and $h-1$ proves (9.12). □

Remark 9.18 (Why the maximum tail is not the square-horizon estimate). The exponent in (9.12) is of order H^2/L. For a general family of disjoint intervals retained by the square horizon one may have $L \asymp x_0^2$ (for M itself such excursions are finitely many by Corollary 9.8), while the relevant witness lies much earlier, at physical time $t \asymp x_0$. Then a violation of size $t^{1/2+\varepsilon}$ need not make H^2/L large. Thus the global maximum tail, although correct, is not the right Catalan quantity for combining with the fractional-square horizon. The correct quantity is the pointwise height at the support time $r_I(t)$.

Lemma 9.19 (Pointwise primitive-Catalan tail)**.** *Let $L = 2n \ge 4$ and let (S_r) be a uniformly chosen primitive Dyck path of length L. For $1 \le r \le L-1$ and $h \ge 1$,*

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 8n^{3/2}\exp\left(-\frac{h^2}{2r}\right) \le 3L^{3/2}\exp\left(-\frac{h^2}{2r}\right). \tag{9.13}$$

By time reversal the same estimate holds with r replaced by $L-r$; hence

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \ge h\} \le 3L^{3/2}\exp\left[-\frac{h^2}{2\min\{r,L-r\}}\right]. \tag{9.14}$$

If $h > \min\{r, L-r\}$ the probability is 0, so the bound is only relevant for $h \le \min\{r, L-r\}$.

Proof. Ignore both positivity and the endpoint constraint. Among all 2^L sign words, the proportion whose first r signs have sum at least h is at most $\exp(-h^2/(2r))$ by the elementary Chernoff bound for a Rademacher sum. Therefore the number of primitive Dyck paths satisfying $S_r \ge h$ is at most

$$2^L \exp\left(-\frac{h^2}{2r}\right).$$

The total number of primitive Dyck paths of length $2n$ is Cat_{n-1}. The standard central-binomial lower bound $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ gives

$$\mathrm{Cat}_{n-1} = \frac{1}{n}\binom{2n-2}{n-1} \ge \frac{4^{n-1}}{2n\sqrt{n-1}}.$$

Since $2^L = 4^n$, division gives the first inequality in (9.13); the second follows from $8n^{3/2} < 3(2n)^{3/2}$. Reversing the path and changing the signs of the reversed increments preserves the primitive Dyck class and carries height S_r to height S_{L-r}, which proves (9.14). □

For an observed excursion I define its pointwise Catalan upper-tail value by

$$p_{\mathrm{Cat}}(I,t) := \mathbb{P}_{\mathrm{pCat},I}\left\{S_{r_I(t)} \ge |M(t)|\right\}. \tag{9.15}$$

This is a tail probability under the fixed-mask Catalan reference law; it is not a probability assigned to the arithmetic Möbius path itself.

Corollary 9.20 (Fractional-square Catalan rarity)**.** *Fix $c > 0$ and $\varepsilon > 0$. Let $I = (x_0, x_1]$ be a complete excursion with $x_1 \le cx_0^2$, and keep its zero mask fixed. For all sufficiently large x_0,*

$$\mathbb{P}_{\mathrm{pCat},I}\left\{\exists\, t \in I :\ S_{r_I(t)} \ge t^{1/2+\varepsilon}\right\} \le 3c^{5/2}x_0^5 \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right). \tag{9.16}$$

In particular the right-hand side is smaller than x_0^{-A} for every fixed $A > 0$ once x_0 is large enough.

Proof. At a support position $t = a_r$ one has $r \le t - x_0 < t$. Hence Lemma 9.19 with $h = t^{1/2+\varepsilon}$ gives

$$\mathbb{P}_{\mathrm{pCat},I}\{S_r \ge t^{1/2+\varepsilon}\} \le 3L^{3/2}\exp\left(-\frac{1}{2}t^{2\varepsilon}\right) \le 3L^{3/2}\exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right).$$

There are at most L support positions. Since $L \le x_1 - x_0 \le x_1 \le cx_0^2$, the union bound gives (9.16). Between support positions the path is constant, so no additional physical times have to be counted. □

The preceding statement is purely combinatorial: it measures how far a large value lies in the tail of a reference law, not how the Möbius constraints select a path. Combined with the localization theorem it gives the following quantitative form.

Corollary 9.21 (Catalan surprise of a violation)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, put $\delta = \Theta - \eta - \frac{1}{2}$, and fix $\lambda > \lambda_*$. Then there are infinitely many excursions $I = (x_0, x_1]$ with $x_1 \le \lambda x_0$ containing a point t with*

$$-\log p_{\mathrm{Cat}}(I,t) \ \ge\ \frac{1}{8}x_0^{2\delta} - O(\log x_0). \tag{9.17}$$

The same holds with $x_1 \le \lambda x_0$ replaced by $x_1 \le cx_0^2$ for any fixed $c > 0$.

Proof. By Corollary 9.7 and Lemma 9.4, at least $(\gamma_1/\pi - 1/\log\lambda - o(1))\log X$ excursions starting at most X have $x_1 \le \lambda x_0$ and contain t with $h = |M(t)| \ge \frac{1}{2}t^{1/2+\delta}$; for cx_0^2 use Lemma 9.2 instead. With $r = r_I(t) < t$ and $L \le x_1 - x_0$, Lemma 9.19 gives $p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp(-h^2/(2r)) \le 3L^{3/2}\exp(-t^{2\delta}/8)$, and $t > x_0$, $L \le \lambda x_0$ (respectively $L \le cx_0^2$). □

Remark 9.22 (Selector bounds are one-sided). Lemma 9.19 gives only the upper-tail estimate

$$p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp\left(-\frac{|M(t)|^2}{2r_I(t)}\right),$$

and hence the one-sided implication

$$-\log p_{\mathrm{Cat}}(I,t) \ge \frac{|M(t)|^2}{2r_I(t)} - \log(3L^{3/2}).$$

No reverse inequality, and therefore no equivalence "up to polynomial factors", is proved here. Consequently a selector bound $-\log p_{\mathrm{Cat}}(I,t) \le \Phi(x_0)$ with $\Phi(x) = x^{o(1)}$ on excursions with $x_1 \le \lambda x_0$ implies $|M(t)| \le t^{1/2}x_0^{o(1)}$ there, so RH follows from Corollary 9.13; this is a sufficient condition, not an independent equivalent formulation. A polynomial lower bound $p_{\mathrm{Cat}} \ge x_0^{-A}$ would similarly give $|M(t)| \ll \sqrt{t\log t}$ on the retained ranges. To turn the Catalan weight into a two-sided measure of deviation one would need a matching lower-tail estimate of the form $-\log p_{\mathrm{Cat}}(I,t) \ll |M(t)|^2/r_I(t) + \log L$, which is not proved here.

Remark 9.23 (The correct quantifier). A single value $|M(t)| \ge t^{1/2+\varepsilon}$, or many values with bounded ratio $|M(t)|/t^{1/2+\varepsilon}$, is compatible with RH, since Littlewood's criterion allows a constant C_ε. The negation of the criterion is

$$\exists\, \varepsilon_0 > 0: \qquad \sup_{t\ge 1}\frac{|M(t)|}{t^{1/2+\varepsilon_0}} = \infty,$$

equivalently there are $t_j \to \infty$ and $A_j \to \infty$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$. Accordingly, the rarity bound (9.16) is a statement about the reference law, and the arithmetic content lies in the sequence form below. If $\Theta > 1/2$ and $0 < \delta < \Theta - \frac{1}{2}$, the oscillation theorem gives $|M(t_j)| \ge \frac{1}{2}t_j^{1/2+\delta}$ along a sequence, and then $|M(t_j)|/t_j^{1/2+\varepsilon} \ge \frac{1}{2}t_j^{\delta-\varepsilon} \to \infty$ for every $0 < \varepsilon < \delta$.

Corollary 9.24 (Catalan cost along a Littlewood-violating sequence)**.** *Let $t_j \to \infty$, $A_j \to \infty$ and $\varepsilon_0 > 0$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$, and let $I_j = (x_{0,j}, x_{1,j}]$ be the excursion containing t_j. If $x_{1,j} \le x_{0,j}^B$ for a fixed B, then*

$$-\log p_{\mathrm{Cat}}(I_j,t_j) \;\ge\; \frac{A_j^2}{2}\,t_j^{2\varepsilon_0} - O_B(\log x_{0,j}) \;\ge\; \frac{A_j^2}{2}\,x_{0,j}^{2\varepsilon_0} - O_B(\log x_{0,j}).$$

Proof. With $r = r_{I_j}(t_j) < t_j$ and $h = |M(t_j)|$, $h^2/(2r) \ge A_j^2 t_j^{1+2\varepsilon_0}/(2t_j)$; apply Lemma 9.19 with $L \le x_{1,j} \le x_{0,j}^B$. □

Selection energies: complete excursions and open prefixes. The support-time clock of Definition 9.16 is the squarefree-compressed clock

$$\mathcal{S}(x) = \#\{n \le x : \mu(n) \ne 0\}, \qquad L^{\#}(I) = \mathcal{S}(x_1) - \mathcal{S}(x_0) \le x_1 - x_0,$$

and $r_I(t) = \mathcal{S}(t) - \mathcal{S}(x_0)$. Two tail quantities are natural: one for the height of a complete excursion, and one for an open prefix, which does not require knowledge of the future return.

Definition 9.25 (Selection energies)**.** For a complete excursion I with height $H(I)$ and compressed length $L^{\#} = L^{\#}(I)$, put

$$p_{\mathrm{Cat}}^{\max}(I) := \mathbb{P}_{\mathrm{pCat},L^{\#}}\{H_{\max} \ge H(I)\}, \qquad \mathcal{I}_{\mathrm{Cat}}(I) := -\log p_{\mathrm{Cat}}^{\max}(I).$$

For $t \geq 1$ with $M(t) \neq 0$ let $a(t)$ be the last zero of M before t and $r = \mathcal{S}(t) - \mathcal{S}(a(t))$. After orientation, the compressed segment from $a(t)$ to t is a positive meander of length r: a ± 1 path $(S_k)_{k \leq r}$ with $S_0 = 0$ and $S_k > 0$ for $1 \leq k \leq r$. Let $\mathcal{M}_r$ be the set of such paths and

$$p_{\mathrm{open}}(t) := \mathbb{P}_{\mathcal{M}_r}\{S_r \geq |M(t)|\},$$

the uniform measure on $\mathcal{M}_r$.

Lemma 9.26 (Height–entropy inequalities)**.** (i) *If $L^{\#} = 2$, the compressed path is $0, 1, 0$ and $H(I) = 1$. For every complete excursion with $L^{\#} \geq 4$,*

$$\mathcal{I}_{\mathrm{Cat}}(I) \;\geq\; \frac{(H(I)-1)^2}{L^{\#}-2} - \log\frac{L^{\#}}{2}.$$

(ii) *$\#\mathcal{M}_r = \binom{r-1}{\lfloor (r-1)/2 \rfloor} \geq 2^r/(4\sqrt{r})$, and for $h \geq 1$*

$$\mathbb{P}_{\mathcal{M}_r}\{S_r \geq h\} \leq 4\sqrt{r}\, e^{-h^2/(2r)}, \qquad -\log p_{\mathrm{open}}(t) \;\geq\; \frac{M(t)^2}{2r} - \log(4\sqrt{r}).$$

Proof. (i) is (9.12). (ii) The count of positive meanders is classical (ballot theorem; the number ending at s is $\frac{s}{r}\binom{r}{(r+s)/2}$). From $\binom{2m}{m} \geq 4^m/(2\sqrt{m})$ and $\binom{2m+1}{m} = \frac{1}{2}\binom{2m+2}{m+1}$ one gets $\#\mathcal{M}_r \geq 2^r/(4\sqrt{r})$. Among all 2^r words the proportion with $S_r \geq h$ is at most $e^{-h^2/(2r)}$ by the Chernoff bound; divide. □

Proposition 9.27 (Selector criteria)**.** (i) Complete excursions. *Fix $\lambda > \lambda_*$. If there is A such that $p_{\mathrm{Cat}}^{\max}(I) \geq x_0^{-A}$ for every sufficiently large complete excursion with $x_1 \leq \lambda x_0$, then $H(I) \ll_{A,\lambda} \sqrt{x_0 \log x_0}$ on that family, and RH holds.*

(ii) Open prefixes. *If there is A such that $p_{\mathrm{open}}(t) \geq t^{-A}$ for all large t with $M(t) \neq 0$, then $|M(t)| \ll_A \sqrt{t \log t}$, and RH holds.*

Proof. (i) If $L^{\#} = 2$ then $H(I) = 1$. Otherwise $L^{\#} \geq 4$ and, by Lemma 9.26(i), $(H-1)^2 \leq (L^{\#}-2)(A \log x_0 + \log L^{\#})$, and $L^{\#} \leq (\lambda - 1)x_0$. Littlewood's bound therefore holds on every excursion with $x_1 \leq \lambda x_0$, and Corollary 9.13(iii), in the form that discards the λ-long excursions, gives RH. (ii) By Lemma 9.26(ii), $M(t)^2 \leq 2r(A \log t + \log(4\sqrt{r}))$ with $r \leq t$. □

Remark 9.28 (Status of the two selectors). Both hypotheses are stronger than RH: each yields $|M| \ll \sqrt{t \log t}$ on its range, beyond any bound known to follow from RH. Neither is known to follow from RH alone: a ballistic stretch of compressed length r has $-\log p \asymp r$, and RH does not exclude such stretches with r a small power of t. The open selector controls every point without reference to the future return; the complete selector uses only the horizon-localized family, where $L^{\#} \leq (\lambda - 1)x_0$ is controlled. By contraposition and Corollary 9.24, failure of RH forces the compressed Möbius path to realize, along an unbounded sequence, sectors of Catalan or meander weight $\exp(-c\, t^{2\delta})$ with $\delta > 0$, not merely polynomially rare ones.

Remark 9.29 (The threshold λ_*). The function $\cos(\gamma_1 \log x)$ changes sign γ_1/π times per unit of $\log x$, i.e. once in every multiplicative interval of ratio $e^{\pi/\gamma_1} = \lambda_*$; the constant in (9.5) is this density. On the geometric side, a chain of λ-long intervals realizes the bound of Lemma 9.4. The method of Theorem 9.11 compares these two counts, and λ_* is the exact point where they balance: for $\lambda \leq \lambda_*$ the λ-long family can, as far as counting alone is concerned, absorb every Kaczorowski–Pintz excursion. Going below λ_* requires information on where inside an excursion the large values occur, not only how many excursions carry them.

Proposition 9.30 (Logarithmic-length budget)**.** *Let $\mathcal{F}$ be a family of excursions and let $u_1 < \cdots < u_N$ be the starts of its members with $2 \leq u_j \leq X$, with ends v_j. The initial excursion $(0, 2]$ is*

explicitly excluded, since $\log(v/u)$ *is not defined at* $u = 0$*. Then*

$$\sum_{j=1}^{N-1} \log \frac{v_j}{u_j} \le \log \frac{X}{2}.$$

If moreover $v_N \le u_N^A$ *for some* $A \ge 1$*, then* $\sum_{j=1}^{N} \log(v_j/u_j) \le A \log X$*.*

Proof. Disjointness gives $u_{j+1} \ge v_j$, so $\log(v_j/u_j) \le \log u_{j+1} - \log u_j$; the sum telescopes to at most $\log u_N - \log u_1 \le \log(X/2)$. The last term is at most $(A-1)\log u_N \le (A-1)\log X$. □

Lemmas 9.1–9.4 are the special cases in which every term is bounded below. The budget form shows what the horizon theorem actually measures.

Corollary 9.31 (Mean logarithmic length of Kaczorowski–Pintz excursions)**.** *Suppose* $\Theta > 1/2$*, fix* $0 < \eta < \Theta - \frac{1}{2}$*, and let* $\mathcal{K}(X)$ *be any set of distinct excursions starting at most* X*, each containing a point* t *with* $|M(t)| \ge \frac{1}{2}t^{\Theta-\eta}$*, such that* $\#\mathcal{K}(X) \ge (\gamma_1/\pi - o(1))\log X$ *(such sets exist by Corollary 9.7). Then*

$$\frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \;\le\; \frac{\pi}{\gamma_1} + o(1) = \log \lambda_* + o(1).$$

Proof. Fix $A > 1$. By Corollary 9.8, all but finitely many excursions satisfy $v \le u^A$. These exceptions form a fixed global finite set (once A is fixed), so the sum of their individual finite values $\log(v/u)$ is $O_A(1)$. Proposition 9.30 gives a total of at most $A \log X + O_A(1)$. Divide by $\#\mathcal{K}(X)$ and let $A \downarrow 1$. □

Remark 9.32 (What lowering λ_* requires). Corollary 9.13(iii) is the statement that the excursions carrying Kaczorowski–Pintz witnesses cannot all have $\log(v/u) > \log\lambda$ with $\lambda > \lambda_*$; Corollary 9.31 is its averaged form. Consequently, any proof that such excursions have mean logarithmic length at least $\pi/\gamma_1 + \delta$ for some fixed $\delta > 0$ would contradict Corollary 9.31 and prove RH, and a surplus $\delta > \log(\lambda_*/\lambda)$ among the excursions whose witness lies beyond λx_0 would extend Corollary 9.13(iii) to that $\lambda < \lambda_*$. Kaczorowski–Pintz, as stated, gives neither: it supplies at least one return in each band $[Y^{1-\delta}, Y]$ but no upper separation between consecutive witnesses. Nor do short-interval estimates for μ supply it: $|M(x+h) - M(x)| = o(h)$ is compatible with M staying positive throughout $(x, x+h]$, whereas a return requires a decrement of size at least $M(x)$ in a prescribed direction. Energy bounds do not help either, since a flat excursion of height 1 has $\beta_I, Q_I \asymp \ell_I$.

Wide peaks from cancellation in all short intervals. The only arithmetic input of the following lemma is the theorem of Matomäki–Teräväinen [13]: for every fixed $\theta > 0.55$, $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ as $x \to \infty$.

Lemma 9.33 (Mesoscopic Lipschitz bound)**.** *Fix* $\Lambda > 1$ *and* $\vartheta > 0.55$*. For large* u *put* $h_u = \lceil (\Lambda u)^\vartheta \rceil$ *and*

$$\eta_u := \max\Big\{ h_u^{-1},\ \max_{u \le a \le \Lambda u} \frac{|M(a+h_u) - M(a)|}{h_u} \Big\}.$$

Then $\eta_u \to 0$*, and for all integers* $u \le a < b \le \Lambda u$*,*

$$|M(b) - M(a)| \le \eta_u (b-a) + h_u. \tag{9.18}$$

Proof. Fix $\theta' \in (0.55, \vartheta)$. For all sufficiently large u, $a + h_u \le 2\Lambda u$ uniformly for $u \le a \le \Lambda u$. The all-interval estimate of [13] therefore yields a function $\varepsilon(u) \to 0$ such that every full block $(b, b + \lceil b^{\theta'} \rceil]$ with $u \le b \le 2\Lambda u$ has Möbius sum at most $\varepsilon(u) b^{\theta'}$ in modulus. Cover $(a, a+h_u]$ by consecutive such blocks and one final piece. The number of full blocks is $O(h_u/u^{\theta'} + 1)$, their total contribution is $O(\varepsilon(u) h_u)$, and the last piece has length $O(u^{\theta'}) = o(h_u)$ because $\theta' < \vartheta$. Hence $|M(a+h_u) - M(a)| = o(h_u)$ uniformly in a, and $\eta_u \to 0$. For (9.18), write $b - a = q h_u + r$ with

$0 \le r < h_u$: the q full blocks start in $[u, \Lambda u]$ and contribute at most $q\eta_u h_u$, and the remainder at most $r < h_u$. □

Proposition 9.34 (Wide peaks)**.** *Let $(u, v]$ be an excursion with $v \le \Lambda u$, let $u < t < v$ with $|M(t)| = H$, and put $A = (H - h_u)_+$ and $m = \lfloor A/\eta_u \rfloor$. Then*

$$t - u \ge \frac{A}{\eta_u}, \qquad v - t \ge \frac{A}{\eta_u},$$

$$\beta \ge H + 2\sum_{j=1}^{m}(A - \eta_u j), \qquad Q \ge H^2 + 2\sum_{j=1}^{m}(A - \eta_u j)^2.$$

Consequently, uniformly for such excursions with $H/h_u \to \infty$,

$$v - u \ge (2 + o(1))\frac{H}{\eta_u}, \qquad \beta \ge (1 + o(1))\frac{H^2}{\eta_u}, \qquad Q \ge \Big(\frac{2}{3} + o(1)\Big)\frac{H^3}{\eta_u},$$

so that $(v - u)/H$, β/H^2 and Q/H^3 tend to infinity, whereas the unit-step bounds of Lemma 3.3 and Proposition 5.7 give only 2, 1 and $\frac{2}{3}$.

Proof. By (9.18) applied to $(u, t]$ and $(t, v]$, $H \le \eta_u(t - u) + h_u$ and $H \le \eta_u(v - t) + h_u$. Thus, since the two distances are integers,

$$t - u,\ v - t \ge \left\lceil \frac{A}{\eta_u} \right\rceil,$$

and therefore $m = \lfloor A/\eta_u \rfloor \le t - u, v - t$. For $1 \le j \le m$ the points $t \pm j$ lie in $[u, v]$, and (9.18) gives $\sigma_\nu M(t \pm j) \ge A - \eta_u j \ge 0$. Summing t and the pairs $t \pm j$ gives the bounds for β and Q. Finally $2\sum_{j\le m}(A - \eta_u j) = A^2/\eta_u + O(A)$, $2\sum_{j\le m}(A - \eta_u j)^2 = \frac{2}{3}A^3/\eta_u + O(A^2)$, and $A = (1 + o(1))H$. □

Corollary 9.35 (Packing of high excursions)**.** *Let $Y = Y(U)$ with $Y/U^\vartheta \to \infty$. The number of excursions with $U \le u \le 2U$, $v \le \Lambda u$ and height at least Y is $o(U/Y)$, whereas the unit-step bound gives $O(U/Y)$.*

Proof. They are disjoint and contained in $[U, 2\Lambda U]$. By Proposition 9.34, each has length at least

$$(2 + o(1))Y/\max_{U\le u\le 2U} \eta_u.$$

□

Remark 9.36 (Scope). Proposition 9.34 is informative only for heights above $u^{0.55+o(1)}$. If $\Theta > 0.55$ and $0.55 < \vartheta < \Theta - \eta$, it applies to the Kaczorowski–Pintz witness excursions of Corollary 9.14 with $v \le \lambda u$: their peaks are wide in the sense above. It does not lower λ_*: the resulting lower bound for $\log(v/u)$ is of order $H/(\eta_u u)$, which may still tend to 0, whereas Remark 9.32 requires a fixed surplus. The published quantitative estimate of Matomäki–Teräväinen [13, Theorem 1.1] makes the gain explicit. If $0.55 < \theta' < \vartheta$, then the blocking argument in Lemma 9.33 gives, for every fixed $\varepsilon > 0$,

$$\eta_u \ll_{\varepsilon,\theta',\vartheta,\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\vartheta}.$$

Consequently, in the regime $H/h_u \to \infty$ of Proposition 9.34,

$$Q \gg_\varepsilon H^3(\log u)^{1/3-\varepsilon}.$$

9.5. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 9.74 is no longer needed: on $(u, 2u]$ the Möbius function is a *linear* expression in the prefix $\mu\mathbf{1}_{[1,u]}$.

Proposition 9.37 (Linear window)**.** *Let $u \geq 1$. For every integer $u < n \leq 2u$,*

$$\mu(n) = -\sum_{\substack{d|n \\ d\leq u}} \mu(d). \tag{9.19}$$

Consequently, if $M(u) = 0$ and $u < t \leq 2u$,

$$M(t) = -\sum_{d\leq u} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{9.20}$$

Proof. Since $n \geq 2$, $\sum_{d|n} \mu(d) = 0$. Every proper divisor of n is at most $n/2 \leq u$, and every divisor $d \leq u$ is proper because $n > u$. Summing (9.19) over $u < n \leq t$, the number of multiples of d in $(u, t]$ is $\lfloor t/d \rfloor - \lfloor u/d \rfloor$. □

Corollary 9.38 (Localized prefix formula)**.** *Let $M(u) = 0$, let $1 < \lambda < 2$, and let $u < t \leq \lambda u$. Then*

$$M(t) = -\sum_{d\leq \lambda u/2} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{9.21}$$

Indeed, if $\lambda u/2 < d \leq u$, then $d > u/2$ and the first multiple of d strictly larger than u is $2d > \lambda u \geq t$.

Corollary 9.39 (Prefix-dependency cutoff)**.** *Put*

$$\theta_* := \frac{1}{2} e^{\pi/\gamma_1} = 0.6244\ldots.$$

For every fixed θ with $\theta_* < \theta < 1$, RH is equivalent to Littlewood's bound on every excursion starting at u throughout

$$u < t \leq \min\{x_1, 2\theta u\},$$

where, throughout that window, $M(t)$ is given by the sum (9.21) using only the prefix $\mu(d)$ with $d \leq \theta u$. In particular, the choice $\theta = 5/8$ corresponds to $\lambda = 5/4$ and gives the RH-equivalent window

$$1 \leq r = t - u \leq \min\{\ell - 1, u/4\}$$

using only $d \leq 5u/8$.

The number θ_* is an infimum, not an attained endpoint, because the horizon theorem requires $\lambda > \lambda_*$.

Proof. Take $\lambda = 2\theta > \lambda_*$ in Corollary 9.13(iii), then use Corollary 9.38. □

Corollary 9.40 (Algebraic dependency gap)**.** *Let $\theta_* < \theta < 1$. If u is a zero of M, then every increment $\mu(n)$ with*

$$u < n \leq 2\theta u$$

is determined by the prefix $\mu(1), \ldots, \mu(\lfloor \theta u \rfloor)$. Consequently the intermediate block

$$(\theta u, u]$$

does not occur as a direct input in the divisor recursion determining the localized future window. As $\theta \downarrow \theta_*$, the omitted block has relative length tending to $1 - \theta_* = 0.3755\ldots$.

Proof. For $u < n \le 2\theta u$, every proper divisor $d \mid n$ satisfies $d \le n/2 \le \theta u$. Hence

$$\mu(n) = -\sum_{\substack{d\mid n\\ d<n}} \mu(d)$$

uses only $d \le \theta u$. □

Remark 9.41 (Direct dependency is not independence). Corollary 9.40 is an algebraic statement about which values occur as inputs of the divisor recursion. It does not assert statistical or arithmetic independence between the omitted block $(\theta u, u]$ and the earlier prefix.

In convolution form, (9.19) says that $(\mu - \mu_u) * 1 = \mu \mathbf{1}_{(u,\infty)}$ on $n \le 2u$, the order-one analogue of (9.32), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (9.20) follows from $\sum_{d\le t} \mu(d)\lfloor t/d \rfloor = 1$, since $\lfloor t/d \rfloor = 1$ for $u < d \le t \le 2u$.

Proposition 9.42 (One-step recursion). *For real $x \ge 1$,*

$$\sum_{k\le x} M\left(\frac{x}{k}\right) = 1. \tag{9.22}$$

Hence, if $M(u) = 0$ and $t > u$,

$$M(t) = -\sum_{2\le k\le t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right)\right], \tag{9.23}$$

with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; hence on a localized window $u < t \le \lambda u$ all arguments are at most $\lambda u/2$. For $\lambda = 5/4$ this is $5u/8$.

Proof. $\sum_{k\le x} M(x/k) = \sum_{k\le x} \sum_{d\le x/k} \mu(d) = \sum_{m\le x} \sum_{d\mid m} \mu(d) = 1$. Subtract the identity at u, where $M(u) = 0$, and note $M(u/k) = 0$ for $k > u$. □

Proposition 9.43 (Scale cascade in the localized excursion window). *Let $(u, v]$ be an excursion, let $\lambda_* < \lambda < 2$, and let $u < t \le \min\{v, \lambda u\}$. For $k \ge 2$ put*

$$\Delta_k(u, t) := M(t/k) - M(u/k),$$

and for an integer $K \ge 2$ put

$$\mathcal{R}_K(u, t) := -\sum_{K<k\le t} \Delta_k(u, t).$$

Then, exactly,

$$M(t) = -\sum_{k=2}^{K} \Delta_k(u, t) + \mathcal{R}_K(u, t), \tag{9.24}$$

and $\mathcal{R}_K(u, t)$ uses only values $M(q)$ with

$$q \le \frac{\lambda u}{K + 1}.$$

Equivalently, all dependence on values above $\lambda u/(K + 1)$ is concentrated in the $K - 1$ increments $\Delta_2, \dots, \Delta_K$. Moreover

$$\Delta_k(u, t) = \sum_{u/k<n\le t/k} \mu(n),$$

so the supporting interval has length $(t - u)/k$.

Proof. Equation (9.24) is Proposition 9.42 with the terms $2 \le k \le K$ separated. If $k > K$, then $t/k \le \lambda u/(K + 1)$ and $u/k \le \lambda u/(K + 1)$. The last identity is the definition of the summatory function on the interval $(u/k, t/k]$. □

Corollary 9.44 (First channels)**.** *For $K = 2$,*

$$M(t) = -\big(M(t/2) - M(u/2)\big) + R_2(u,t),$$

where R_2 uses only $q \le \lambda u/3$. With the convenient localized choice $\lambda = 5/4$, the only channel above $5u/12$ is

$$\Delta_2(u, u+r) = \sum_{u/2 < n \le u/2 + r/2} \mu(n), \qquad 0 < r \le u/4,$$

whose supporting interval has length at most $u/8$. In the limiting sense $\lambda \downarrow \lambda_*$, the successive remainder cutoffs are

$$\frac{\lambda_*}{2}u = 0.6244\ldots u, \quad \frac{\lambda_*}{3}u = 0.4162\ldots u, \quad \frac{\lambda_*}{4}u = 0.3122\ldots u, \quad \frac{\lambda_*}{5}u = 0.2497\ldots u.$$

Lemma 9.45 (Channel persistence extracted from a high excursion)**.** *Let $S_r = \sigma_\nu M(u+r) > 0$ on $1 \le r \le h$ with $S_0 = 0$ and $|S_r - S_{r-1}| \le 1$. Fix $K \ge 2$ and write, using Proposition 9.43,*

$$S_r = \sum_{k=2}^{K} X_k(r) + Y_K(r), \quad X_k(r) := -\sigma_\nu \Delta_k(u, u+r), \quad Y_K(r) := \sigma_\nu \mathcal{R}_K(u, u+r).$$

If $H = \max_{r \le h} S_r$ and τ is a point with $S_\tau = H$, then among the last $\lfloor H/2 \rfloor + 1$ indices

$$\tau - \lfloor H/2 \rfloor, \ldots, \tau$$

there is one fixed component among $X_2, \ldots, X_K, Y_K$ which is at least $H/(2K)$ at no fewer than $\lfloor H/(2K) \rfloor$ indices.

Proof. Since $S_0 = 0$ and the path has unit steps, $\tau \ge H$ and $S_{\tau - j} \ge H - j \ge H/2$ for $0 \le j \le \lfloor H/2 \rfloor$. At every one of these indices the sum of the K displayed components is at least $H/2$, hence some component is at least $H/(2K)$. Pigeonholing the indices among the K components gives the claim. □

Remark 9.46 (What the persistence lemma adds). Lemma 9.45 is not a bound for M. It converts a large value attained inside one excursion into a quantified persistence requirement: either one of the finitely many scaled interval sums $\Delta_2, \ldots, \Delta_K$ repeatedly carries a positive fraction of the height, or the responsibility lies repeatedly in the deeper remainder, which is supported entirely below $\lambda u/(K+1)$. This avoids the iteration defect of Remark 9.60: no new point is claimed to be the start-window point of another excursion.

Corollary 9.47 (Channel descent with explicit loss)**.** *Let u be a zero of M, $t > u$, $2 \le K \le u$, and $0 < \alpha < 1$. Then either*

(i) *there are $2 \le k \le K$ and $q \in \{\lfloor u/k \rfloor, \lfloor t/k \rfloor\}$ with $|M(q)| \ge |M(t)|/(2K)$, hence*

$$\frac{|M(q)|}{q^\alpha} \ge \frac{k^\alpha}{2K} \cdot \frac{|M(t)|}{t^\alpha}, \qquad q \le \left\lfloor \frac{t}{2} \right\rfloor;$$

or

(ii) *$|\mathcal{R}_K(u,t)| \ge |M(t)|/K$, where $\mathcal{R}_K$ depends only on $M(q)$ with $q \le t/(K+1)$.*

The statement holds at every $t > u$, not only in a localized window.

Proof. The proof of Proposition 9.43 applies verbatim to every $t > u$, with $\mathcal{R}_K$ supported on $q \le t/(K+1)$. By (9.24), one of the K terms $\Delta_2, \ldots, \Delta_K, \mathcal{R}_K$ has modulus at least $|M(t)|/K$. If it is $\Delta_k = M(\lfloor t/k \rfloor) - M(\lfloor u/k \rfloor)$, one of its two values has modulus at least $|M(t)|/(2K)$, and, since $k \ge 2$, $q \le \lfloor t/2 \rfloor$. □

Remark 9.48 (Channels do not iterate to a bound). Corollary 9.47 has no iteration gap: it applies at the new point q relative to the last zero of M before q. Its defect is the constant loss $k^\alpha/(2K) < 1$ per step. If alternative (i) held at every step, a chain of length $m \asymp \log t$ would give only $|M(t)| \le$

$(2K)^m\,O(1) = t^{O(\log K)}$, weaker than the trivial bound. For fixed K and $0 < \alpha < 1$ the guaranteed factor can never reach 1, since

$$\frac{k^\alpha}{2K} \le \frac{K^\alpha}{2K} = \frac{1}{2K^{1-\alpha}} < \frac{1}{2}.$$

Thus no iteration of the same pigeonhole estimate can remove the loss at an individual step. A closing argument must instead recover the loss collectively from the persistence supplied by Lemma 9.45, or prove cancellation in the deep remainder. Both require information beyond dependency alone.

Proposition 9.49 (Quantified persistence along channel descent)**.** *Fix $K \ge 2$, $0 < \alpha < 1$ and $\delta > 0$. Put*

$$c_{\alpha,K} := \log\!\left(\frac{2K}{2^\alpha}\right) > 0.$$

Suppose $t_0 \to \infty$ through points satisfying

$$Z_\alpha(t_0) := \frac{|M(t_0)|}{t_0^\alpha} \ge t_0^\delta.$$

Starting from t_0, iterate Corollary 9.47 relative to the last zero before the current point, as long as alternative (i) *occurs. If no deep-remainder alternative occurs during the first m steps, then there are integers $2 \le k_j \le K$ and points t_{j+1} such that*

$$t_{j+1} \le \left\lfloor \frac{t_j}{2} \right\rfloor, \qquad Z_\alpha(t_{j+1}) \ge \frac{k_j^\alpha}{2K} Z_\alpha(t_j),$$

and hence

$$Z_\alpha(t_m) \ge Z_\alpha(t_0) \prod_{j<m} \frac{k_j^\alpha}{2K} \ge t_0^\delta e^{-mc_{\alpha,K}}. \tag{9.25}$$

Consequently, with

$$m_* := \left\lfloor \frac{\delta \log t_0}{2c_{\alpha,K}} \right\rfloor, \qquad \eta_{\alpha,K,\delta} := \frac{\delta \log 2}{2c_{\alpha,K}},$$

either a deep-remainder alternative occurs before step m_*, or for all sufficiently large t_0 there is a descendant t_{m_*} satisfying

$$t_{m_*} \le t_0^{1-\eta_{\alpha,K,\delta}+o(1)}, \qquad Z_\alpha(t_{m_*}) \ge t_0^{\delta/2}.$$

Proof. At each application of alternative (i), the integer bound in Corollary 9.47 gives $t_{j+1} \le \lfloor t_j/2 \rfloor$. The normalized lower bound multiplies by $k_j^\alpha/(2K)$, proving the first inequality in (9.25). Since $k_j \ge 2$, each factor is at least $2^\alpha/(2K) = e^{-c_{\alpha,K}}$, which proves the second. For $m \le m_*$ the right side is at least $t_0^{\delta/2}$. Also $t_m \le t_0/2^m$, which at $m = m_*$ gives the stated power reduction. If at some stage the last zero before t_j is $< K$, then t_j lies in one of the finitely many excursions starting below K and is bounded independently of t_0; this is incompatible with $Z_\alpha(t_j) \ge t_0^{\delta/2}$ for large t_0. Thus, unless the deep remainder has already occurred, the iteration is valid through m_*. □

Remark 9.50 (What remains after quantified persistence). Proposition 9.49 shows that a polynomial normalized violation cannot disappear immediately under channel descent. It does not yield a contradiction: the pointwise loss accumulates, and bounding the deep remainder by absolute values loses a factor of the number of channels. A closing argument must therefore exploit the multi-index persistence of Lemma 9.45 collectively, or establish cancellation among the channels entering $\mathcal{R}_K$.

Remark 9.51 (What the recursion determines). Iterating (9.22) determines all of M from $M(1) = 1$, so determinacy alone carries no information; cf. Remark 11.2. The content of (9.23), after Corollary 9.39, is more precise: at the start u of an excursion, every value in an RH-equivalent window $u < t \le \lambda u$ is a signed sum of increments of M over scaled intervals $(u/k, t/k]$ lying below $\lambda u/2$. This is a descent of dependency only; no descent of normalized size follows from the identity.

Corollary 9.52 (RH in the localized linear window)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\left|\sum_{d\le 5u/8}\mu(d)\left(\left\lfloor\frac{t}{d}\right\rfloor-\left\lfloor\frac{u}{d}\right\rfloor\right)\right| = \left|\sum_{2\le k\le t}\left[M\left(\frac{t}{k}\right)-M\left(\frac{u}{k}\right)\right]\right| \ll_\varepsilon t^{1/2+\varepsilon}$$

for every excursion starting at u and every

$$u < t \le \min\left\{x_1, \frac{5}{4}u\right\}.$$

Equivalently, in intra-excursion coordinates $t = u + r$ it is enough to control

$$1 \le r \le \min\{\ell - 1, u/4\}.$$

Proof. Use Corollary 9.39 with $\theta = 5/8$, together with Proposition 9.42. □

Proposition 9.53 (Kernel form of the recursion)**.** *Let $M(u) = 0$ and let t be an integer with $u < t < 2u$. For $1 \le q \le t/2$ put*

$$c_q(t,u) = \left(\left\lfloor\frac{t}{q}\right\rfloor-\left\lfloor\frac{t}{q+1}\right\rfloor\right)-\left(\left\lfloor\frac{u}{q}\right\rfloor-\left\lfloor\frac{u}{q+1}\right\rfloor\right).$$

Then:

(i) $M(t) = -\sum_{1\le q\le t/2} c_q(t,u)M(q)$;

(ii) $c_q(t,u) = \dfrac{t-u}{q(q+1)} + e_q$ *with* $|e_q| < 2$; *for* $q > \sqrt{t}$ *one has* $c_q \in \{-1,0,1\}$, *and* $c_q \ne 0$ *only if* $q = \lfloor t/k\rfloor$ *or* $q = \lfloor u/k\rfloor$ *for some* $k < \sqrt{t}$;

(iii) $c_1(t,u) \ge \frac{1}{2}(t-u-1)$;

(iv) $M(t) = -(t-u)\,\Upsilon(\sqrt{t}) + \mathrm{Err}(t,u)$, *where* Υ *is the coefficient* (12.10) *and*

$$|\mathrm{Err}(t,u)| \le 2\sum_{q\le\sqrt{t}}|M(q)| + \sum_{k<\sqrt{t}}\left(\left|M\left(\frac{t}{k}\right)\right|+\left|M\left(\frac{u}{k}\right)\right|\right).$$

Proof. (i) In (9.23) group the indices $k \ge 2$ by the value $q = \lfloor t/k\rfloor$, respectively $q = \lfloor u/k\rfloor$. The number of $k \ge 1$ with $\lfloor t/k\rfloor = q$ is $\lfloor t/q\rfloor - \lfloor t/(q+1)\rfloor$, and for $q \le t/2$ none of them is $k = 1$. The same holds for u, because $k = 1$ gives $q = u > t/2$. (ii) Each bracket equals $x/(q(q+1))$ minus a difference of two fractional parts, so it differs from $x/(q(q+1))$ by less than 1. For $q > \sqrt{t} \ge \sqrt{u}$ both $t/q - t/(q+1)$ and $u/q - u/(q+1)$ are < 1, so each bracket is 0 or 1, and it is 1 only if $q = \lfloor x/k\rfloor$ with $k \le x/q < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2\rfloor - \lfloor u/2\rfloor)$ and $\lfloor t/2\rfloor - \lfloor u/2\rfloor \le \lceil (t-u)/2\rceil$. (iv) Insert (ii) into (i) for $q \le \sqrt{t}$; the smooth part is $(t-u)\sum_{q\le\sqrt{t}} M(q)/(q(q+1)) = (t-u)\Upsilon(\sqrt{t})$. For $q > \sqrt{t}$ use the last assertion of (ii). □

Corollary 9.54 (Loss of the pointwise induction)**.** *Let $0 < \alpha < 1$ and suppose $|M(q)| \le Cq^\alpha$ for all $q \le t/2$. Then, for $u < t < 2u$ with $M(u) = 0$,*

$$\left|M(t) + (t-u)\Upsilon(\sqrt{t})\right| \ll_\alpha C\,t^{(1+\alpha)/2}, \tag{9.26}$$

while the termwise bound of the full kernel gives only $|M(t)| \ll_\alpha C\,t$. For $\alpha = \frac{1}{2} + \varepsilon$ the pointwise induction therefore loses $t^{1/2-\varepsilon}$, and $t^{1/4-\varepsilon/2}$ after the smooth component $-(t-u)\Upsilon(\sqrt{t})$ is extracted.

Proof. By Proposition 9.53(iv), $|\mathrm{Err}| \le 2C\sum_{q\le\sqrt{t}} q^\alpha + 2Ct^\alpha\sum_{k<\sqrt{t}} k^{-\alpha} \ll_\alpha Ct^{(1+\alpha)/2}$. The full kernel satisfies $\sum_q |c_q|q^\alpha \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_q |c_q|q^\alpha \ll_\alpha t$ by (ii). □

The smooth component is not small for termwise reasons: by (12.12),

$$\Upsilon(\sqrt{t}) = -\sum_{n>\sqrt{t}}\frac{M(n)}{n(n+1)},$$

so its size is governed by the global identity $\sum_{n\geq 1} M(n)/(n(n+1)) = 0$, i.e. by the pole of ζ at $s = 1$ (Remark 12.14), and involves values of M beyond $t/2$. Even under RH the tail gives only $(t-u)\Upsilon(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $\mathrm{Err}(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

Remark 9.55 (Localized intra-excursion system). Let $I = (u, x_1]$ be an excursion of length $\ell = x_1 - u$ and orientation σ_ν, and put

$$S_r := \sigma_\nu M(u+r) > 0 \qquad (1 \leq r < \ell), \qquad S_0 = S_\ell = 0.$$

For a fixed $\lambda \in (\lambda_*, 2)$ set

$$h_\lambda(u) := \min\{\ell - 1, (\lambda - 1)u\}.$$

Then the entire RH-relevant initial segment satisfies simultaneously

$$S_r = -\sigma_\nu \sum_{q \leq \lambda u/2} c_q(u+r, u) M(q) > 0, \qquad 1 \leq r \leq h_\lambda(u),$$

and $|S_r - S_{r-1}| \leq 1$. For $\lambda = 5/4$ this is the segment $r \leq \min\{\ell - 1, u/4\}$ and all dependencies lie below $5u/8$.

Definition 9.56 (Relaxed divisor-sum window class)**.** Fix $u \geq 2$ and $1 \leq h < u$. For $\delta = (\delta_d)_{d \leq u} \in \{-1, 0, 1\}^u$ with $\delta_1 = 1$ and $\sum_{d \leq u} \delta_d = 0$, put

$$D_\delta(n) := \sum_{\substack{d \mid n \\ d \leq u}} \delta_d, \qquad u < n \leq u + h.$$

We call δ window-admissible with orientation $\sigma_\nu \in \{-1, 1\}$ if $D_\delta(n) \in \{-1, 0, 1\}$ throughout the window and

$$\sigma_\nu \sum_{u < n \leq u+r} D_\delta(n) < 0 \qquad (1 \leq r \leq h).$$

The induced oriented path is $S_r = -\sigma_\nu \sum_{u<n\leq u+r} D_\delta(n) > 0$. For the genuine Möbius prefix, $\delta_d = \mu(d)$, these are exactly the increment and excursion conditions coming from $\mu * 1 = \delta_1$.

Proposition 9.57 (Ballistic counterexample to the relaxed window class)**.** *The relaxed class of Definition 9.56 admits ballistic paths. Namely, take*

$$\delta_1 = 1, \qquad \delta_u = -1, \qquad \delta_d = 0 \quad (1 < d < u).$$

Then for every $u < n < 2u$ one has $D_\delta(n) = 1$, and with $\sigma_\nu = -1$,

$$S_r = r \qquad (1 \leq r < u).$$

If $m_q := \sum_{d \leq q} \delta_d$, then $m_q = 1$ for every $q < u$. Hence for any $0 < \alpha < 1$, any fixed $1 < \lambda < 2$, and $h = \lfloor (\lambda - 1)u \rfloor$,

$$\sup_{q \leq \lambda u/2} \frac{|m_q|}{q^\alpha} = 1, \qquad \frac{S_h}{(u+h)^\alpha} \asymp_\lambda u^{1-\alpha} \longrightarrow \infty.$$

Thus a maximum principle comparing the normalized localized-window height to the normalized history is false on this relaxed class.

Proof. For $u < n < 2u$, the integer u does not divide n. Hence among the two nonzero coordinates of δ only $d = 1$ contributes to $D_\delta(n)$, so $D_\delta(n) = 1$. The remaining assertions are immediate. □

Remark 9.58 (What the relaxation loses). The counterexample violates the internal divisor identities of the Möbius prefix. Already for every prime $p < u$ the genuine identity requires $\delta_1 + \delta_p = 0$, whereas the ballistic vector above has $\delta_p = 0$. Consequently any successful localized argument of this type must use some of the relations

$$\sum_{d \mid n} \delta_d = 0 \qquad (2 \leq n \leq u),$$

not merely the future-window constraints.

Remark 9.59 (The prefix correlation identity). The complete-excursion bridge identity must not be applied to a truncated localized window. For $1 \le h < \ell$, put $x_j = \sigma_\nu \mu(u+j)$, $S_r = \sum_{j \le r} x_j$, and $C_a^{[h]} = \sum_{j=1}^{h-a} x_j x_{j+a}$. Lemma 5.1 applied to the prefix of length h gives exactly

$$\sum_{r=1}^{h-1} S_r^2 + \sum_{a=1}^{h-1} a C_a^{[h]} = S_h \sum_{r=1}^{h-1} S_r.$$

The right-hand side vanishes only for a complete bridge. Thus $Q_\nu = -\sum_a a C_a^{(\nu)}$ cannot be used as an identity internal to a truncated horizon unless the excursion actually returns there. In particular, passing from a large localized value to the energy of the whole excursion mixes the retained window with the part deliberately discarded by the horizon theorem.

Remark 9.60 (The one-step iteration gap). Even a one-step implication sending a large value in the localized window to a large value at some $q \le \lambda u/2$ would not automatically iterate. The point q need not lie in the localized window of the excursion that contains q; it may lie beyond $\lambda u'$ relative to that excursion's start u'. Sparsity of the family of λ-long excursions does not repair a single descent chain. Iteration therefore requires either a principle valid for all t, or a mechanism that places the new point inside an admissible window at every step.

Proposition 9.61 (Termwise absolute-value contraction gives no nontrivial gain)**.** *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_q = \sum_{r \le h} w_r c_q(u+r, u)$, so that $\sum_r w_r M(u+r) = -\sum_q K_q M(q)$. If, for some $0 < \alpha < 1$ and $C_\alpha > 0$,*

$$\sum_q |K_q| q^\alpha \le C_\alpha \sum_{r \le h} w_r (u+r)^\alpha,$$

then

$$\sum_{r \le h} w_r |M(u+r)| \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u+r)^\alpha.$$

Proof. Proposition 9.53(iii) gives $c_1(u+r, u) \ge (r-1)/2 \ge 0$, hence $\frac{1}{2}\sum_r w_r(r-1) \le K_1 \le C_\alpha \sum_r w_r (u+r)^\alpha$. Since $M(u) = 0$ and M has unit steps, $|M(u+r)| \le r$, and the stated inequality follows. □

Remark 9.62 (No strict coercivity from the first coefficient). Writing $t = u + r$ gives the exact formula

$$c_1(u+r, u) = \begin{cases} \lceil r/2 \rceil, & u \text{ even}, \\ \lfloor r/2 \rfloor, & u \text{ odd}. \end{cases}$$

In particular $c_1(u+1, u) = 0$ when u is odd. The preceding proposition therefore rules out only this particular termwise absolute-value mechanism; it does not rule out a global signed contraction or cancellation among the coefficients K_q.

Definition 9.63 (Multiplicative layers of the prefix)**.** Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity. For $R \ge 2$ and $k \ge 1$, a vector $(\delta_d)_{d \le R}$ satisfies the first k Möbius layers if $\delta_1 = 1$ and

$$\sum_{d \mid n} \delta_d = 0 \qquad (2 \le n \le R,\ \Omega(n) \le k).$$

Thus layer 1 imposes the identities at primes, layer 2 adds prime squares and products of two primes, and so on.

Proposition 9.64 (One-layer propagation into the localized future)**.** *Suppose $(\delta_d)_{d\le R}$ satisfies the first k Möbius layers. Then*

$$\delta_d = \mu(d) \qquad (d \le R,\ \Omega(d) \le k).$$

Moreover, for every integer n with

$$R < n \le 2R, \qquad \Omega(n) \le k+1,$$

one has

$$-\sum_{\substack{d\mid n\\ d\le R}} \delta_d = \mu(n).$$

Thus the first k multiplicative layers of the prefix determine the future increments at every integer n in the stated window $R < n \le 2R$ with $\Omega(n) \le k+1$. No converse statement about all future increments is asserted.

Proof. The first assertion follows recursively from $\delta_n = -\sum_{d\mid n,\ d<n} \delta_d$: every proper divisor of an integer with $\Omega(n) \le k$ has smaller Ω. For the second, every proper divisor of $n \le 2R$ is at most $n/2 \le R$ and has $\Omega(d) \le k$. Hence the displayed sum equals $-\sum_{d\mid n,\ d<n} \mu(d) = \mu(n)$. □

Question 9.65 (Multiplicative-layer depth)**.** Fix $\theta \in (\theta_*, 1)$ and the localized window $u < n \le 2\theta u$. Let $k = k(u)$ be a depth up to which the internal Möbius divisor identities are imposed below $R = \theta u$. What growth of $k(u)$, if any, suffices to combine those identities with $M(u) = 0$ and the unit-step and excursion-sign constraints so as to force $\max_{u<t\le 2\theta u} |M(t)| \ll_\varepsilon u^{1/2+\varepsilon}$? A bounded depth would give a compressed arithmetic mechanism; imposing all layers reconstructs the Möbius prefix. Proposition 9.64 makes the depth a quantifiable intermediate invariant; the results below treat only fixed k.

Proposition 9.66 (Layer sums in the localized window)**.** *Fix $1 < \lambda < 2$ and an integer $j \ge 1$ before letting $u \to \infty$. Then*

$$\sum_{\substack{u<n\le \lambda u\\ \Omega(n)=j}} \mu(n) = (-1)^j\,(1+o(1))\,\frac{(\lambda-1)\,u\,(\log\log u)^{j-1}}{(j-1)!\,\log u}. \tag{9.27}$$

Proof. If $\Omega(n) = j$ and $\mu(n) \ne 0$, then n is squarefree with exactly j prime factors and $\mu(n) = (-1)^j$. By Landau's theorem (see [14, Ch. 7]), the number $\pi_j^\flat(x)$ of such $n \le x$ satisfies $\pi_j^\flat(x) \sim f_j(x) := x(\log\log x)^{j-1}/((j-1)!\log x)$. Since $f_j(x)/x$ is slowly varying, $f_j(\lambda u) - f_j(u) = (\lambda - 1 + o(1))f_j(u)$. □

Corollary 9.67 (Bounded layer depth)**.** *Fix $k \ge 1$, $\theta \in (\theta_*, 1)$, $R = \theta u$, and the window $u < n \le 2\theta u$.*

(i) *The increments fixed by the first k layers (Proposition 9.64), namely those with $\Omega(n) \le k+1$, form a set of $o(u)$ integers of the window.*

(ii) *Their summed contribution over the window is*

$$\sum_{\substack{u<n\le 2\theta u\\ \Omega(n)\le k+1}} \mu(n) = (-1)^{k+1}(1+o(1))\frac{(2\theta-1)\,u\,(\log\log u)^k}{k!\,\log u},$$

so it exceeds $u^{1-\varepsilon}$ in modulus and has a fixed sign.

(iii) *Let $\delta_d = \mu(d)$ for $\Omega(d) \le k$ and $\delta_d = 0$ otherwise. For squarefree n in the window with $\omega(n) = w \ge k+2$,*

$$D_\delta(n) = \sum_{\substack{d\mid n\\ d\le R}} \delta_d = \sum_{j=0}^{k} (-1)^j \binom{w}{j} = (-1)^k \binom{w-1}{k},$$

which has modulus at least $k+1 \geq 2$. Thus the zero extension of a k-layer prefix violates the unit-step constraint at every such n, i.e. at all but $o(u)$ squarefree integers of the window.

Proof. (i) and (ii) follow from Proposition 9.66 summed over $j \leq k+1$, the term $j = k+1$ dominating. For (iii), all divisors of n other than n are $\leq n/2 \leq R$, and n itself has $\Omega(n) = w > k$, so $D_\delta(n)$ is the truncated alternating binomial sum, evaluated by induction on k via $\binom{w-1}{k} = \binom{w}{k} - \binom{w-1}{k-1}$. The proportion of squarefree $n \leq x$ with $\omega(n) \leq k+1$ tends to 0 by Proposition 9.66. □

Remark 9.68 (What fixed layer depth proves). For every fixed k, Corollary 9.67 shows that the first k layers determine only a density-zero subset of the localized window, while the signed sum of the determined increments has modulus $(1+o(1))(2\theta-1)u(\log\log u)^k/(k!\log u)$, larger than $u^{1-\varepsilon}$ for all large u. Hence, if the full sum on the window is $O(u^{1/2+\varepsilon})$, substantial cancellation must occur between the increments determined by any fixed set of initial layers and the remaining ones. The results proved here do not determine how the required depth grows with u; in particular they do not show that $\asymp \log\log u$ layers (the typical size of Ω) are necessary or sufficient, which would require estimates uniform in $k = k(u)$. Part (iii) shows that the unit-step constraint is far from vacuous on a k-layer relaxation: it forbids the naive completion at almost every integer of the window, so the free coordinates δ_d with $\Omega(d) > k$ are strongly constrained.

9.6. **Conditional extreme-value consequence of excursion sparsity.** The counts of Section 9.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\Omega, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \leq x_{0,k+1}$, and real random variables Z_k (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_\flat$; let $k_\mathcal{F}(1) < k_\mathcal{F}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \leq X\} \qquad (\max\emptyset = -\infty).$$

The lemmas of Section 9.1 hold for every ω.

Hypothesis 9.69 (SG_ϖ)**.** There are $C, \kappa_0, \varpi > 0$ such that for every $j \geq 1$ and $z \geq 0$

$$\Pr(k_\mathcal{F}(j) \text{ exists and } Z_{k_\mathcal{F}(j)} > z) \leq Ce^{-\kappa_0 z^\varpi}.$$

For $\varpi = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

Theorem 9.70 (Extreme scale from a count)**.** *Assume* (SG_ϖ) *for $\mathcal{F}$ and suppose that $E_\mathcal{F}(X) \leq \Lambda(X)$ for every ω, with $\Lambda(X) \geq 1$ deterministic. Then*

$$\Pr(Z_{\max,\mathcal{F}}(X) > z) \leq \Lambda(X)\, Ce^{-\kappa_0 z^\varpi}. \tag{9.28}$$

Consequently, for every $\delta > 0$,

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{\kappa_0}\log\Lambda(X)\Big)^{1/\varpi}\Big) \leq C\,\Lambda(X)^{-\delta}, \tag{9.29}$$

and $Z_{\max,\mathcal{F}}(X) = O_{\Pr}\big((\log\Lambda(X))^{1/\varpi}\big)$ as $\Lambda(X) \to \infty$.

Proof. The members of $\mathcal{F}$ starting at most X are the first $E_\mathcal{F}(X) \leq \Lambda(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{j \leq \Lambda(X)} \{k_\mathcal{F}(j) \text{ exists},\ Z_{k_\mathcal{F}(j)} > z\}$, and (9.28) is the union bound. Substituting $\kappa_0 z^\varpi = (1+\delta)\log\Lambda(X)$ gives (9.29). □

Corollary 9.71 (Triple-logarithmic scale on the square family)**.** *Assume (SG) for the square-exceptional family $\mathcal{F}_{x^2}$ of the model. Then, for $X \geq e^e$ and every $\delta > 0$,*

$$\Pr(Z_{\max,\square}(X) > z) \leq 4C(\log\log X)e^{-\kappa_0 z^2}, \tag{9.30}$$

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{\kappa_0}\log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{9.31}$$

Thus $Z_{\max,\square}(X) = O_{\Pr}(\sqrt{\log\log\log X})$. The same holds for the family $\mathcal{F}_{cx^2}$ of Lemma 9.2 with a constant depending on c.

Proof. Use (9.2) in (9.28) and note that $e^{-\kappa_0 z^2} = (\log\log X)^{-(1+\delta)}$ at the stated z. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to Λ gives an almost sure bound with the same constant.

Theorem 9.72 (Almost sure extreme scale)**.** *Assume* (SG_ϖ) *for $\mathcal{F}$ and $E_{\mathcal{F}}(X) \le \Lambda(X)$ for every ω, where Λ is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X\to\infty} \frac{Z_{\max,\mathcal{F}}(X)}{(\log\Lambda(X))^{1/\varpi}} \le \kappa_0^{-1/\varpi}.$$

Proof. Fix $\delta > 0$. For $j \ge j_0$ let X_j be the least X with $\Lambda(X) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/\kappa_0\big)^{1/\varpi}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (9.28), $\Pr(A_j) \le 2^{j+1}Ce^{-(1+\delta)(j+1)\log 2} = C2^{-(j+1)\delta}$, which is summable, so almost surely A_j occurs only finitely often. For $X_j \le X \le X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in X gives $Z_{\max,\mathcal{F}}(X) \le z_j$ outside A_j, while $\log\Lambda(X) \ge j\log 2$. Hence the lim sup is at most $\lim_j\big((1+\delta)(j+1)/(\kappa_0 j)\big)^{1/\varpi} = ((1+\delta)/\kappa_0)^{1/\varpi}$. Let $\delta \downarrow 0$ along a countable sequence. □

With $\Lambda_\square(X) = 2 + \log_2(\log X/\log 2)$, which is continuous and increasing, Theorem 9.72 gives under (SG)

$$\limsup_{X\to\infty} \frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \le \kappa_0^{-1/2} \quad \text{almost surely.}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most $X+1$; Lemma 9.4; Lemma 9.1), Theorems 9.70 and 9.72 give three levels:

family	members starting $\le X$	extreme scale under (SG)
all excursions	$\le X+1$	$\sqrt{\log X}$
$x_1 > \lambda x_0$	$O(\log X)$	$\sqrt{\log\log X}$
$x_1 > cx_0^2$ (or x_0^A)	$O(\log\log X)$	$\sqrt{\log\log\log X}$

Each level of lacunarity removes one logarithm from the extreme scale. Under (SG_ϖ) the last column reads $(\log X)^{1/\varpi}$, $(\log\log X)^{1/\varpi}$, $(\log\log\log X)^{1/\varpi}$.

Remark 9.73 (Scope). For M itself the square-exceptional family is finite by Corollary 9.8, so Theorems 9.70–9.72 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 9.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

9.7. **The square range as an exact bilinear form of the prefix.**

Proposition 9.74 (Bilinear square-range identity)**.** *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor. \tag{9.32}$$

In particular, if $M(u) = 0$ and $u < t \le u^2$,

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{9.33}$$

Proof. Let $\mu_u = \mu \mathbf{1}_{[1,u]}$. Each factor of $(\mu - \mu_u) * (\mu - \mu_u)$ is supported on integers $\geq u+1$, so the convolution, and its convolution with 1, vanish on $n < (u+1)^2$, in particular on $n \leq u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \leq u^2$; sum over $n \leq t$. □

Remark 9.75 (Higher-order determinacy). For every $k \geq 1$ and $n \leq u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (\mu_u^{*j} * 1^{*(j-1)})(n). \tag{9.34}$$

Indeed, $(\mu - \mu_u)^{*k}$ is supported on integers at least $(u+1)^k$, so after convolution with $1^{*(k-1)}$ it still vanishes for $n < (u+1)^k$, in particular for $n \leq u^k$. Expanding gives

$$0 = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \mu^{*(k-j)} * \mu_u^{*j} * 1^{*(k-1)}.$$

For $j = 0$, $\mu^{*k} * 1^{*(k-1)} = \mu$. For $j \geq 1$, pair $k-j$ factors of μ with $k-j$ of the factors 1 to obtain $\mu^{*(k-j)} * 1^{*(k-j)} = \delta$; the remaining $j-1$ factors of 1 produce the term in (9.34). Rearranging proves the formula. Thus determination by the prefix is not special to u^2. What is special to $t \leq u^2$ is the order-two form (9.33), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj) \rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [4]; summed forms of this kind are used in the fast computation of M [5].

Question 9.76 (Square-excursion bridges)**.** Let $u = x_{0,\nu}$ and $x_{1,\nu} \leq u^2$. Then $\mathcal{B}_u(u) = \mathcal{B}_u(x_{1,\nu}) = 0$ and $-\sigma_\nu \mathcal{B}_u(t) > 0$ for $u < t < x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj) \rfloor$ and these bridge and sign constraints be combined to prove

$$\max_{u < t \leq x_{1,\nu}} |\mathcal{B}_u(t)| \ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon}$$

uniformly over square-ordinary excursions? By Corollary 9.12 this uniform bound is equivalent to RH. By Corollary 9.52 it suffices to consider $t \leq \min\{x_{1,\nu}, \frac{5}{4}u\}$, where $\mathcal{B}_u(t)$ reduces to the localized linear form and uses only $\mu(d)$ with $d \leq 5u/8$.

9.8. **Mellin mass of a complete excursion.**

Proposition 9.77 (Excursion blocks of $1/\zeta$)**.** *For a complete excursion $I = (x_0, x_1]$ put $D_I(s) = \sum_{x_0 < n \leq x_1} \mu(n) n^{-s}$. Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{9.35}$$

For real $\varsigma > 0$, $D_I(\varsigma)$ has the sign of the excursion and

$$|D_I(\varsigma)| \geq \varsigma \beta_I x_1^{-\varsigma-1} \geq \varsigma H_I^2 x_1^{-\varsigma-1}. \tag{9.36}$$

For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{9.37}$$

zero runs contributing nothing.

Proof. For $a_n = \mu(n)$, $A(n) = M(n)$ and $f(n) = n^{-s}$, finite summation by parts gives

$$\sum_{n=x_0+1}^{x_1} a_n f(n) = A(x_1)f(x_1) - A(x_0)f(x_0) + \sum_{n=x_0+1}^{x_1-1} A(n)\big(f(n) - f(n+1)\big).$$

Since $M(x_0) = M(x_1) = 0$, this is (9.35). For real $\varsigma > 0$, all interior $M(n)$ have the excursion sign, and

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} x^{-\varsigma-1}\, dx \geq \varsigma x_1^{-\varsigma-1}.$$

The bound $\beta_I \geq H_I^2$ is (5.14). If M vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every n with $\mu(n) \neq 0$ lies in a completed excursion. For $\Re s > 1$ the ordinary Dirichlet series is absolutely convergent, so summing the finite disjoint blocks gives (9.37). □

Theorem 9.78 (Abscissa of absolute excursion-block convergence)**.** *Let $D_I(s)$ be the completed-excursion blocks of Proposition 9.77. The series*

$$\sum_I D_I(s)$$

has abscissa of absolute convergence, with respect to the excursion blocks, exactly

$$\boxed{\sigma_{a,\mathrm{exc}} = \Theta}.$$

More precisely, it converges absolutely and locally uniformly for $\Re s > \Theta$ and there represents $1/\zeta(s)$, while for every real $0 < \varsigma < \Theta$ one has

$$\sum_I |D_I(\varsigma)| = \infty.$$

Consequently,

$$\mathrm{RH} \iff \sum_I D_I(s) \text{ converges absolutely for every } \Re s > \tfrac{1}{2}.$$

This is block absolute convergence; the ordinary termwise Dirichlet series $\sum_n \mu(n) n^{-s}$ still has abscissa of absolute convergence 1*.*

Proof. Let $s = \varsigma + i\tau$. Since

$$n^{-s} - (n+1)^{-s} = s \int_n^{n+1} x^{-s-1}\, dx,$$

with the real positive branch of $\log x$, one has

$$|n^{-s} - (n+1)^{-s}| \leq |s| n^{-\varsigma-1}.$$

Hence

$$\sum_I |D_I(s)| \leq |s| \sum_{n\geq 1} |M(n)| n^{-\varsigma-1}.$$

Let $K_0 \Subset \{\Re s > \Theta\}$ be compact and set $\varsigma_0 = \min_{s\in K_0} \Re s > \Theta$. Choose $0 < \epsilon_0 < \varsigma_0 - \Theta$. From $B_1(X) = X^{1+\Theta+o(1)}$ we have $B_1(X) \ll_{\epsilon_0} X^{1+\Theta+\epsilon_0}$ for large X, and partial summation gives

$$\sum_{n\geq 1} |M(n)| n^{-\varsigma_0-1} < \infty.$$

Since $\sup_{s\in K_0} |s| < \infty$, the Weierstrass criterion gives normal convergence of the block series on K_0. Thus it is holomorphic in $\Re s > \Theta$. There are no zeros of ζ in that half-plane by the definition of Θ; at $s = 1$ the function $1/\zeta(s)$ has a removable zero. After filling that point, $1/\zeta$ is holomorphic throughout the connected half-plane. The identity already holds for $\Re s > 1$, so the identity theorem extends it to all $\Re s > \Theta$.

For real $\varsigma > 0$, order the completed excursions by their right endpoints. If V is a zero of M, the fixed sign of each excursion and finite Abel summation give

$$\sum_{I:\, x_1(I)\leq V} |D_I(\varsigma)| = \sum_{n<V} |M(n)| (n^{-\varsigma} - (n+1)^{-\varsigma}).$$

Letting $V \to \infty$ through zeros and using monotone convergence gives

$$\sum_I |D_I(\varsigma)| = \sum_{n\ge 1} |M(n)|\big(n^{-\varsigma} - (n+1)^{-\varsigma}\big), \tag{9.38}$$

even when both sides are infinite. If the right side were finite for some $0 < \varsigma < \Theta$, then for $n \le X$

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} t^{-\varsigma-1}\, dt \ge \varsigma (X+1)^{-\varsigma-1},$$

so

$$B_1(X) \le \frac{(X+1)^{\varsigma+1}}{\varsigma} \sum_{n\ge 1} |M(n)|\big(n^{-\varsigma} - (n+1)^{-\varsigma}\big) = O(X^{1+\varsigma}),$$

contradicting $B_1(X) = X^{1+\Theta+o(1)}$. Therefore the block abscissa is exactly Θ. □

Remark 9.79 (Status of the result). The input $D_M(X) = X^{\Theta+o(1)}$ is Pintz's theorem; the new point here is the excursion grouping. The fixed sign of each completed excursion turns Abel summation into the positive identity (9.38), so the rightmost-zero exponent becomes an absolute-convergence boundary for the grouped reciprocal-zeta series. We did not locate this excursion-block formulation in the literature.

10. Two-point correlations and the energy of excursions

For integers $0 \le u < X$ and $1 \le h < X - u$ put

$$C_h^{(u,X]} := \sum_{u<n\le X-h} \mu(n)\mu(n+h), \qquad C_h^{[X]} := C_h^{(0,X]}.$$

For an excursion $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ one has $C_h^{(\nu)} = C_h^{(x_{0,\nu},x_{1,\nu}]}$, since $\sigma_\nu^2 = 1$. Chowla's conjecture asserts $C_h^{[X]} = o(X)$ for each fixed $h \ge 1$; it is open. Matomäki, Radziwiłł and Tao [11] prove, for $H = H(X) \to \infty$, $H \le X$,

$$\sum_{h\le H} \left| \sum_{n\le X} \mu(n)\mu(n+h) \right| = o(HX).$$

Since replacing the upper limit X by $X - h$ changes the h-th sum by at most h, we shall use the triangular consequence

$$\sum_{h\le H} |C_h^{[X]}| = o(HX) \qquad (H \to \infty,\ H = o(X)). \tag{10.1}$$

Their stronger averaged Elliott form permits translated shifts of size $O(X)$ and covers μ.

10.1. **Signed lag averages and the weighted RH criterion.**

Remark 10.1 (Unweighted signed averages are degenerate). By Lemma 5.4, every excursion satisfies $\sum_h C_h^{(\nu)} = -q_\nu/2$, with $q_\nu \le \ell_\nu$. Hence $\ell_\nu^{-2} \sum_h C_h^{(\nu)} = O(\ell_\nu^{-1})$ for purely algebraic reasons: a signed average over lags of the correlations inside a bridge is small without any pseudorandomness of μ. Statements of Chowla type inside an excursion must therefore carry absolute values inside the lag average, or impose a genuinely weighted condition.

The weighted sum behaves differently. By (5.5), $Q_\nu = -\sum_h h C_h^{(\nu)}$: the weight h records at which lags the correlation responsible for the excursion energy sits. The same identity holds globally at every zero of M, and the cross terms between different excursions cancel exactly.

Proposition 10.2 (Weighted correlations at zeros of M)**.** *Let $v \geq 2$ be a zero of M. Then*

$$\sum_{n<v} M(n)^2 = -\sum_{h<v} h\, C_h^{[v]}, \qquad \sum_{h<v} h\, C_h^{[v]} = \sum_{\nu:\, x_{1,\nu}\leq v} \sum_{h} h\, C_h^{(\nu)}. \tag{10.2}$$

Thus, in the h-weighted sum, the correlations between increments lying in different completed excursions, or in zero runs, cancel exactly.

Proof. Apply Lemma 5.1 to the block $x_n = \mu(n)$, $1 \leq n \leq v$: then $S_t = M(t)$ and $S_v = M(v) = 0$, which gives the first identity. Because $M(v) = 0$, $\sum_{n\leq v} M(n)^2 = \sum_{n<v} M(n)^2$. By (5.9), this equals $\sum_\nu Q_\nu$ over the completed excursions ending at or before v, and $Q_\nu = -\sum_h hC_h^{(\nu)}$ by (5.5). □

Theorem 10.3 (Weighted-correlation criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every zero } v \text{ of } M.$$

Proof. If RH holds, Theorem 8.6 and (10.2) give the bound. Conversely, fix a target $\varepsilon_0 > 0$ and choose $\varepsilon > 0$ so small that $(1+\varepsilon)(2+\varepsilon) \leq 2+\varepsilon_0$. Let x be large and put $N = \lfloor x \rfloor$. If $M(N) = 0$, then $\sum_{n\leq x} M(n)^2 = B_2(N)$ is already evaluated at a zero. Otherwise, let $u = \max\{n \leq N : M(n) = 0\}$ and let $v > N$ be the next zero. Then $(u, v]$ is the completed excursion whose open prefix contains N, and Corollary 9.8 gives $v \leq u^{1+\varepsilon} \leq x^{1+\varepsilon}$ for all sufficiently large x. In either case, with $v = N$ in the zero case,

$$\sum_{n\leq x} M(n)^2 \leq \sum_{n<v} M(n)^2 \ll_\varepsilon v^{2+\varepsilon} \leq x^{2+\varepsilon_0}.$$

Theorem 8.6 now applies. □

Proposition 10.4 (Square-root lag cutoff at zeros)**.** *Let $v \to \infty$ through zeros of M, and let $H = H(v) \to \infty$ with $H \leq v^{1/2}$. Then*

$$\sum_{h\leq H} h\, C_h^{[v]} = o(H^2 v). \tag{10.3}$$

In particular,

$$\sum_{n<v} M(n)^2 = -\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} + o(v^2), \tag{10.4}$$

where the lower limit may be replaced by $\lfloor \sqrt{v} \rfloor$ without changing the statement.

Proof. By (10.1), $\sum_{h\leq H} |C_h^{[v]}| = o(Hv)$. Therefore

$$\left| \sum_{h\leq H} hC_h^{[v]} \right| \leq H \sum_{h\leq H} |C_h^{[v]}| = o(H^2 v).$$

Taking $H = \lfloor \sqrt{v} \rfloor$ and using Proposition 10.2 gives (10.4). □

Corollary 10.5 (RH after deleting all small lags)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every sufficiently large zero } v \text{ of } M.$$

Thus the lags $h \leq \sqrt{v}$ are unconditionally negligible at the critical v^2 scale of the weighted-correlation criterion.

Theorem 10.6 (Zero-edge lag localization)**.** *Let $v \to \infty$ through zeros of M. Then*

$$-\sum_{h<v} hC_h^{[v]} = \sum_{n<v} M(n)^2 = v^{1+2\Theta+o(1)}. \tag{10.5}$$

Moreover, for every fixed $0 \le \alpha < \Theta$,

$$\sum_{h\le v^\alpha} hC_h^{[v]} = o\left(\sum_{n<v} M(n)^2\right), \tag{10.6}$$

and hence

$$-\sum_{v^\alpha<h<v} hC_h^{[v]} = (1+o(1))\sum_{n<v} M(n)^2. \tag{10.7}$$

Thus the weighted correlation mass is forced beyond every power v^α with $\alpha < \Theta$.

Proof. The first equality is Proposition 10.2, and the asymptotic exponent is Corollary 8.11. For $H = \lfloor v^\alpha \rfloor$ with $\alpha > 0$, (10.1) gives

$$\left|\sum_{h\le H} hC_h^{[v]}\right| \le H \sum_{h\le H} |C_h^{[v]}| = o(H^2 v) = o(v^{1+2\alpha}).$$

Choose a fixed δ_0 with $0 < \delta_0 < 2(\Theta - \alpha)$. Along zeros $v \to \infty$, Corollary 8.11 gives $B_2(v) \ge v^{1+2\Theta-\delta_0}$ for all sufficiently large v, while the small-lag contribution is $o(v^{1+2\alpha})$. Their ratio is therefore $o(v^{-2(\Theta-\alpha)+\delta_0}) = o(1)$. The case $\alpha = 0$ follows trivially from the fixed finite number of lags and $|C_h^{[v]}| \le v$. □

Remark 10.7 (Endpoint versus the universal square-root cutoff)*.* Proposition 10.4 is uniform without knowing Θ and reaches the endpoint $H = \sqrt{v}$ at the absolute v^2 scale. Theorem 10.6 is stronger in relative terms but requires a strict power gap $\alpha < \Theta$; it does not settle the endpoint $\alpha = \Theta$. Under RH it says that every range $h \le v^{1/2-\delta}$ carries $o(1)$ of the total weighted mass, while if RH fails the negligible range extends correspondingly beyond the square-root scale.

Remark 10.8 (Where the cancellation must occur)*.* The trivial bound is $\sum_h h|C_h^{[v]}| \le \sum_h h(v-h) \le v^3/6$. RH requires the signed weighted sum to be $v^{2+o(1)}$. Absolute values cannot give this at the natural scale: even square-root cancellation $|C_h^{[v]}| \ll v^{1/2+o(1)}$ in every single lag would give only $\sum_h h|C_h^{[v]}| \ll v^{5/2+o(1)}$. Corollary 10.5 sharpens the location of the unresolved cancellation: at the v^2 scale, only lags $h > \sqrt{v}$ remain. By Proposition 10.2 the signed weighted identity itself decomposes excursion by excursion.

10.2. **Lag localization of the energy of an excursion.**

Proposition 10.9 (Energy beyond a lag cutoff)**.** *For every excursion I_ν of length $\ell = \ell_\nu$ and every $1 \le h_0 < \ell$,*

$$\Big|Q_\nu + \sum_{h_0<h<\ell} h\, C_h^{(\nu)}\Big| \le h_0 \sum_{h\le h_0} |C_h^{(\nu)}|.$$

Consequently, if $\sum_{h\le h_0} |C_h^{(\nu)}| \le \delta h_0 \ell$, then

$$\sum_{h_0<h<\ell} h\, C_h^{(\nu)} = -Q_\nu + O(\delta h_0^2 \ell).$$

Proof. Split $Q_\nu = -\sum_{h<\ell} hC_h^{(\nu)}$ at h_0. □

Corollary 10.10 (Averaged Chowla on linear-scale intervals)**.** *Let $(u_j, u_j + L_j]$ be intervals with $L_j \to \infty$ and $u_j \le BL_j$ for a fixed B, and let $h_j \to \infty$ with $h_j = o(L_j)$. Then*

$$\sum_{h \le h_j} |C_h^{(u_j, u_j+L_j]}| = o(h_j L_j).$$

Proof. This is the translated averaged Elliott form of [11]; equivalently, one may subtract the two initial sums at $u_j + L_j$ and u_j and absorb the $O(h)$ boundary terms. Since $u_j = O(L_j)$ and $h_j = o(L_j)$, all translated parameters remain on the scale permitted by that theorem. □

Corollary 10.11 (Large-lag concentration in linear-scale excursions)**.** *Let I_j be excursions with $\ell_j \to \infty$, $x_{0,j} \le B\ell_j$ and heights H_j with $H_j^3/\ell_j \to \infty$. For every $h_j \to \infty$ with $h_j^2 \ell_j = o(H_j^3)$,*

$$\sum_{h_j < h < \ell_j} h\, C_h^{(I_j)} = -(1 + o(1)) Q_{I_j}.$$

If in addition $\Theta > 1/2$, there is a fixed $\Lambda > 1$ with $x_{1,j} \le \Lambda x_{0,j}$, and these excursions carry Kaczorowski–Pintz witnesses $H_j \ge \frac{1}{2} x_{0,j}^{\Theta-\eta}$, then for every fixed

$$0 < \rho < \frac{3(\Theta - \eta) - 1}{2}$$

one may take $h_j = x_{0,j}^{\rho}$, and hence

$$\sum_{x_{0,j}^{\rho} < h < \ell_j} h\, C_h^{(I_j)} = -(1 + o(1)) Q_{I_j}.$$

Proof. By Corollary 10.10, $\sum_{h \le h_j} |C_h^{(I_j)}| = \delta_j h_j \ell_j$ with $\delta_j \to 0$. Proposition 10.9 gives an error at most $\delta_j h_j^2 \ell_j = o(H_j^3) = o(Q_{I_j})$. In the witness case the hypotheses $x_{0,j} \le B\ell_j$ and $x_{1,j} \le \Lambda x_{0,j}$ give $\ell_j \asymp x_{0,j}$. Hence $H_j^3/\ell_j \gg x_{0,j}^{3\Theta-3\eta-1}$, so the displayed condition on ρ gives $h_j^2 \ell_j = o(H_j^3)$. □

Remark 10.12 (Direction of the lag statement). Corollary 10.11 says that the energy lies beyond every cutoff h_j satisfying $h_j = o(H_j^{3/2} \ell_j^{-1/2})$; it does *not* say that the energy is concentrated only at lags larger than the endpoint scale $H_j^{3/2} \ell_j^{-1/2}$ itself.

10.3. **Uniform two-point cancellation in sublinear intervals.** The linear-scale restriction can be removed for mesoscopic lags by a Fourier-uniform estimate valid in every interval in the range displayed below. The needed input is Theorem 1.5 and Remark 5.2 of Matomäki–Teräväinen [13]: for every fixed $0 < A < 1$ and $\eta > 0$,

$$\sup_{\alpha \in \mathbb{R}} \Big| \sum_{x < n \le x+L} \mu(n) e(\alpha n) \Big| \ll_{A,\eta} \frac{L}{(\log x)^A} \qquad (x^{3/5+\eta} \le L \le x). \tag{10.8}$$

The enlargement from the stated $A < 1/3$ to $A < 1$ in their theorem is recorded explicitly in [12].

Theorem 10.13 (Uniform two-point Cesàro cancellation)**.** *Fix $0 < A < 1$ and $\eta > 0$. Uniformly for all integers u, L, R with $u^{3/5+\eta} \le L \le u$ and $1 \le R \le L/2$,*

$$\frac{1}{R} \sum_{h=1}^{R} \frac{|C_h^{(u,u+L]}|}{L - h} \ll_{A,\eta} \Big(\frac{L}{R}\Big)^{1/2} (\log u)^{-A}. \tag{10.9}$$

In particular, if $R \ge L(\log u)^{-2A+2\kappa}$ with $0 < \kappa < A$, the left side is $O((\log u)^{-\kappa})$.

Proof. Put $a_i = \mu(u+i)$ for $1 \le i \le L$ and $A_I(\alpha) = \sum_i a_i e(i\alpha)$. For $h > 0$ define $C_{-h}^{(u,u+L]} := \sum_{i=1}^{L-h} a_{i+h}a_i = C_h^{(u,u+L]}$ and put $C_0 = q = \sum_i a_i^2$. These are the Fourier coefficients of $|A_I(\alpha)|^2$. Parseval gives the precise identity

$$\int_0^1 |A_I(\alpha)|^4\, d\alpha = q^2 + 2\sum_{h=1}^{L-1} |C_h^{(u,u+L]}|^2 \le \|A_I\|_\infty^2 q \le L\|A_I\|_\infty^2.$$

Hence Cauchy–Schwarz gives $\sum_{h\le R}|C_h| \le \sqrt{RL}\,\|A_I\|_\infty$, and since $L - h \ge L/2$ the left side of (10.9) is at most $2\|A_I\|_\infty/\sqrt{RL}$. Apply (10.8). □

Corollary 10.14 (Near-log-squared mesoscopic range)**.** *Fix $\eta > 0$ and $0 < \delta < 1$. Uniformly for*

$$u^{3/5+\eta} \le L \le u, \qquad \frac{L}{(\log u)^{2-\delta}} \le R \le \frac{L}{2},$$

one has

$$\frac{1}{R}\sum_{h\le R} \frac{|C_h^{(u,u+L]}|}{L-h} \ll_{\eta,\delta} (\log u)^{-\delta/4}.$$

Proof. In Theorem 10.13 take $A = 1 - \delta/4$ and $\kappa = \delta/4$. □

Remark 10.15 (What is and is not new). Theorem 10.13 is a short Parseval–Cauchy–Schwarz corollary of the published Fourier estimate (10.8); it does not use the excursion condition. Its role here is to provide an arithmetic input on every adaptively selected interval, something that "almost all intervals" results cannot supply. The factor $(L/R)^{1/2}$ is inherent in this direct L^2-to-L^1 Cauchy–Schwarz step using only the displayed second-moment control of the C_h; additional information on the distribution of the correlations could in principle improve it.

Corollary 10.16 (Sublinear large-lag estimate)**.** *Let $I = (u, u+L]$ be a complete excursion with $u^{3/5+\eta} \le L \le u$, and fix $0 < \delta < 1$. For all sufficiently large u, put*

$$R = \left\lceil \frac{L}{(\log u)^{2-\delta}} \right\rceil.$$

Then

$$\sum_{R<h<L} hC_h^{(I)} = -Q_I + O_{\eta,\delta}\left(\frac{L^3}{(\log u)^{4-7\delta/4}}\right). \tag{10.10}$$

Consequently, if $Q_I L^{-3}(\log u)^{4-7\delta/4} \to \infty$, then the left side is $-(1+o(1))Q_I$.

Proof. Corollary 10.14 gives $\sum_{h\le R}|C_h^{(I)}| \ll RL(\log u)^{-\delta/4}$. Hence

$$\left|\sum_{h\le R} hC_h^{(I)}\right| \le R\sum_{h\le R}|C_h^{(I)}| \ll R^2 L(\log u)^{-\delta/4},$$

which is the error in (10.10) after inserting the chosen value of R into $Q_I = -\sum_{h<L} hC_h^{(I)}$. □

Remark 10.17 (Connection with wide peaks). The quantitative form of the short-interval estimate in [13, Theorem 1.1] gives, after the blocking argument of Lemma 9.33, $\eta_u \ll_\varepsilon (\log u)^{-1/3+\varepsilon} + u^{-c}$ for some $c > 0$ (depending on the two fixed short-interval exponents used there). Thus, in the regime $H/h_u \to \infty$ of Proposition 9.34,

$$Q_I \gg_\varepsilon H^3(\log u)^{1/3-\varepsilon}.$$

Combining this with Corollary 10.16, a sufficient condition for large-lag concentration is

$$\left(\frac{H}{L}\right)^3 (\log u)^{13/3-7\delta/4-\varepsilon} \longrightarrow \infty.$$

For linearly localized Kaczorowski–Pintz excursions satisfying $v \le \lambda u$ with a fixed $\lambda < 2$, this machinery applies when $\Theta > 3/5$ (after choosing the loss parameters so that $\Theta - \eta > 3/5 + \eta'$), but it still does not yield a contradiction: the excursion may be much longer than its height.

10.4. **Fejér reduction and the weaker local-variance target.**

Remark 10.18 (Excursion geometry alone does not give cancellation). For the tent excursion $a_i = +1$ $(i \le k)$, $a_i = -1$ $(k < i \le 2k)$, which satisfies the bridge and one-sign constraints of a primitive excursion, $C_h = 2k - 3h$ for $1 \le h < k$. Hence for $R \to \infty$, $R = o(k)$,

$$\frac{1}{R} \sum_{h \le R} \frac{|C_h|}{2k-h} \longrightarrow 1, \qquad \sum_{h<R} (R-h) C_h \asymp kR^2,$$

whereas $RL \asymp kR$. Arithmetic input is indispensable.

Lemma 10.19 (Finite Fejér identity). *For a real sequence $(a_i)_{1 \le i \le L}$, extended by 0 outside $[1, L]$, put*

$$q = \sum_{i=1}^{L} a_i^2, \qquad C_h = \sum_{i=1}^{L-h} a_i a_{i+h}, \qquad F_R := \sum_{h=1}^{R-1} (R-h) C_h.$$

Then for $1 \le R \le L$,

$$Rq + 2F_R = \sum_{m \in \mathbb{Z}} \left(\sum_{j=1}^{R} a_{m+j} \right)^2. \tag{10.11}$$

Proof. Expand the square. Each diagonal term occurs R times, and a pair at lag h, $1 \le h < R$, occurs $2(R-h)$ times. □

Proposition 10.20 (Bridge–Fejér identity and interior variance). *Let $I = (u, u+L]$ be a complete excursion, put $a_i = \mu(u+i)$ and $S_t = \sum_{i=1}^{t} a_i = M(u+t)$ for $0 \le t \le L$. Then $S_0 = S_L = 0$, and for every $1 \le R < L$,*

$$Rq + 2F_R = 2Q_I - 2 \sum_{t=1}^{L-1-R} S_t S_{t+R}. \tag{10.12}$$

Moreover

$$Rq + 2F_R = E_{\mathrm{bd}}(R) + \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m) \big)^2, \tag{10.13}$$

where

$$E_{\mathrm{bd}}(R) = \sum_{t=1}^{R-1} S_t^2 + \sum_{t=L-R+1}^{L-1} S_t^2 = O(R^3).$$

Consequently, if $R = o(\sqrt{L})$, then

$$F_R = o(RL) \quad \Longleftrightarrow \quad \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m) \big)^2 = Rq + o(RL). \tag{10.14}$$

If in addition $L \ge u^{1/2+\varepsilon}$ and $L \le u$, then

$$q = \left(\frac{6}{\pi^2} + o(1) \right) L, \tag{10.15}$$

and the right side of (10.14) *is equivalently*

$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m) \big)^2 = \left(\frac{6}{\pi^2} + o(1) \right) RL. \tag{10.16}$$

Proof. Extend S_t by 0 outside $0 \le t \le L$. Since $\sum_{j=1}^{R} a_{m+j} = S_{m+R} - S_m$, (10.11) gives

$$Rq + 2F_R = \sum_m (S_{m+R} - S_m)^2 = 2\sum_m S_m^2 - 2\sum_m S_m S_{m+R},$$

which is (10.12). Splitting the same sum into $m < 0$, $0 \le m \le L - R$, and $m > L - R$ gives (10.13): for $-R < m < 0$ the contribution is S_{m+R}^2, for $L - R < m < L$ it is S_m^2, and the endpoint terms vanish because $S_0 = S_L = 0$. Since $|a_i| \le 1$ and the path returns to zero, $|S_t| \le t$ and $|S_{L-t}| \le t$, whence $E_{\mathrm{bd}}(R) \le 2\sum_{t<R} t^2 = O(R^3)$. Thus $E_{\mathrm{bd}}(R) = o(RL)$ when $R = o(\sqrt{L})$, proving the equivalence. Finally the classical estimate $\sum_{n \le x} \mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ gives $q = (6/\pi^2)L + O(\sqrt{u+L})$. Under $u^{1/2+\varepsilon} \le L \le u$,

$$\frac{\sqrt{u+L}}{L} \le \frac{\sqrt{2u}}{u^{1/2+\varepsilon}} \ll u^{-\varepsilon},$$

so the error is uniform and (10.15) follows. □

Remark 10.21 (Fejér density at frequency zero). With $A_I(\alpha) = \sum_{i=1}^{L} a_i e(i\alpha)$ and

$$K_R(\alpha) = \frac{1}{R}\left|\sum_{j=1}^{R} e(j\alpha)\right|^2,$$

one has

$$q + \frac{2F_R}{R} = \int_0^1 |A_I(\alpha)|^2 K_R(\alpha)\, d\alpha.$$

Thus $F_R = o(RL)$ says that the Fejér average of the local spectral density at frequency 0, at resolution $1/R$, equals $q + o(L)$. It is not a statement that $|A_I(\alpha)|^2$ is spectrally flat on the whole circle.

Remark 10.22 (Three distinct targets). The following conditions must not be conflated.

First, absolute averaged Chowla, $\sum_{h<R} |C_h| = o(RL)$, implies only $|F_R| \le R\sum_{h<R} |C_h| = o(R^2 L)$, not the sharper $F_R = o(RL)$ when $R \to \infty$. Conversely, $F_R = o(RL)$ allows cancellation between the C_h and does not control their absolute values.

Second, F_R does not by itself control the small-lag contribution to the excursion energy. If

$$V_R := \sum_{h<R} hC_h,$$

then exactly

$$F_R + V_R = R\sum_{h<R} C_h. \tag{10.17}$$

Thus a Fejér estimate alone does not imply $V_R = o(Q_I)$.

Third, Proposition 10.20 requires both endpoints of the block to be zeros of M. For a block beginning at a zero but ending before the return, additional boundary terms remain.

Remark 10.23 (Why the variance target may go below 3/5). For the Fejér problem one does not need the stronger absolute estimate $\sum |C_h|$. By Proposition 10.20, on a complete excursion and for $R = o(\sqrt{L})$ it is enough to prove the diagonal local-variance law

$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL).$$

This is strictly weaker than absolute averaged Chowla and is therefore a plausible target below the Fourier-uniform threshold $L \ge u^{3/5+\eta}$. The all-interval U^2/Fourier estimates currently used above do not, by themselves, supply this asymptotic for every adaptively selected block in the range

$L \geq u^{1/2+\varepsilon}$. Results holding for almost all starting points cannot be substituted, because excursion endpoints are selected by M itself.

No fixed-prefix formula for the increments in (10.16) is asserted here. In particular, the identity $\sum_{d\leq x}\mu(d)\lfloor x/d\rfloor = 1$ introduces lower-scale terms such as $M(x/2)$ when one truncates at $x/2$; omitting those terms would give an invalid reduction.

11. Trial-division horizon of an excursion

Lemma 11.1 (Square horizon)**.** *Let $u \geq 2$. Trial division by the primes $p \leq u$ alone determines $\mu(n)$ for every $n \leq u^2$.*

Proof. Remove the prime factors $p \leq u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \leq u^2$ has all prime factors $> u$, so it is 1 or a single prime. □

Remark 11.2 (Determinacy is not information). Primes $> u$ are not logically new information, and determination by a finite prefix is not special to u^2: (9.34) propagates a prefix to every power range, and (9.22) determines M from $M(1)$. The ranges $(u, 2u]$ and $(u, u^2]$ are special only in the concrete sense that the exact summatory formula is of order one ((9.20)) or two ((9.33)) in the prefix, and that one-stage trial division by primes $\leq u$ suffices on $(u, u^2]$.

Definition 11.3 (Smooth and rough parts)**.** For $y > 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\leq y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 11.4. *$\mu(n) = \mu(s_y(n))\,\mu(r_y(n))$. If $n < y^k$ then $\Omega(r_y(n)) \leq k-1$, and if $n \leq y^2$ then $r_y(n)$ is 1 or a prime.*

Proof. The two parts are coprime. If $\Omega(r) \geq 1$ then $y^{\Omega(r)} < r \leq n$. □

Remark 11.5 (The horizon u^2 is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu} \leq u_\nu$. Every $n \leq v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, below u_ν. The regime $t > u_\nu^2$ of Section 12 lies beyond v_ν for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 13 gives the local version inside an excursion.

12. Large-prime decomposition at level U^2

The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [14, 22].

Definition 12.1 (Smooth baseline and large-prime part)**.** For $U \geq 2$ and $t \geq 1$ put

$$M_U^{\mathrm{sm}}(t) := \sum_{\substack{n\leq t\\ P^+(n)\leq U^2}} \mu(n), \qquad M_U^{\mathrm{lp}}(t) := \sum_{\substack{n\leq t\\ P^+(n)> U^2}} \mu(n), \qquad M(t) = M_U^{\mathrm{sm}}(t) + M_U^{\mathrm{lp}}(t). \tag{12.1}$$

Proposition 12.2 (Single-prime copy)**.** *For a prime p and $t < p^2$,*

$$\Delta_p(t) := \sum_{a\leq t/p} \mu(ap) = -M(t/p). \tag{12.2}$$

Proof. $a \leq t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 12.3 (Sign content of one prime cone)**.** *Let $\mathcal{S}(y) = \sum_{a\leq y}\mu(a)^2$. Among the numbers ap with $a \leq y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y\rfloor - \mathcal{S}(y), \tag{12.3}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proof. $\mu(ap) = -\mu(a)$, and $\#\{a \le y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

Proposition 12.4 (First-order window)**.** *If $U^2 < t < U^4$, no $n \le t$ has two prime factors $> U^2$ (with multiplicity), and*

$$M_U^{\mathrm{lp}}(t) = -\sum_{U^2 < p \le t} M(t/p). \tag{12.4}$$

Proof. Two such factors would give $n > U^4 > t$. Hence $M_U^{\mathrm{lp}}(t) = \sum_{U^2<p\le t} \Delta_p(t)$, and $t < U^4 < p^2$; apply (12.2). □

Proposition 12.5 (Prime-count form)**.** *For $U^2 < t < U^4$ and $R := t/U^2$,*

$$M_U^{\mathrm{lp}}(t) = -\sum_{a \le R} \mu(a)(\pi(t/a) - \pi(U^2)), \qquad M(t) = M_U^{\mathrm{sm}}(t) - \sum_{a \le R} \mu(a)\pi(t/a) + M(R)\pi(U^2). \tag{12.5}$$

Proof. Expand (12.4) as $-\sum_{U^2<p\le t} \sum_{a \le t/p} \mu(a)$. This is a finite sum, so reversing the order is purely combinatorial. For fixed a the prime runs over $U^2 < p \le t/a$, which is non-empty only if $a < R$. The displayed sum may harmlessly be written with $a \le R$: if R is an integer, the endpoint $a = R$ contributes $\pi(U^2) - \pi(U^2) = 0$. □

Thus the only input from primes $> U^2$ is the value of π at the points t/a, $a \le R$, weighted by $\mu(a)$ at the small scale $R < U^2$.

Definition 12.6 (Copy multiplicities)**.** For $U^2 < t < U^4$, $R = t/U^2$ and $1 \le y \le \lfloor R \rfloor$ put

$$N_y(t,U) = \#\Big\{p : \max(U^2, \tfrac{t}{y+1}) < p \le \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad M_U^{\mathrm{lp}}(t) = -\sum_{1 \le y \le \lfloor R \rfloor} N_y(t,U) M(y). \tag{12.6}$$

Remark 12.7 (Asymptotics of the multiplicities). For fixed $R > 1$ and $U \to \infty$,

$$N_y \sim \frac{RU^2}{2\log U} \cdot \frac{1}{y(y+1)} \quad (1 \le y \le \lfloor R \rfloor - 1), \qquad N_{\lfloor R \rfloor} \sim \frac{U^2}{2\log U}\Big(\frac{R}{\lfloor R \rfloor} - 1\Big) \quad (R \notin \mathbb{Z}), \tag{12.7}$$

and $N_R = 0$ if $R \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor R \rfloor + 1) < U^2$.

Definition 12.8 (Coherent coefficient)**.**

$$\mathcal{C}(R) := \int_1^R \frac{M(w)}{w^2}\, dw = \sum_{a \le R} \mu(a)\Big(\frac{1}{a} - \frac{1}{R}\Big) = \sum_{a \le R} \frac{\mu(a)}{a} - \frac{M(R)}{R}. \tag{12.8}$$

Proposition 12.9 (Fixed R)**.** *For fixed $R > 1$, as $U \to \infty$,*

$$M_U^{\mathrm{lp}}(RU^2) = -\frac{RU^2}{2\log U}(\mathcal{C}(R) + o(1)). \tag{12.9}$$

Proof. In (12.5) the sum is finite; for $a < R$ the prime number theorem gives $\pi(RU^2/a) - \pi(U^2) \sim (R/a - 1)U^2/(2\log U)$, and $\sum_{a \le R} \mu(a)(R/a - 1) = R\mathcal{C}(R)$. □

Lemma 12.10 (The multiplicity-weighted coefficient)**.** *The coefficient*

$$\Upsilon(R) := \sum_{1 \le y \le R} \frac{M(y)}{y(y+1)} = \sum_{n \le R} \frac{\mu(n)}{n} - \frac{M(R)}{\lfloor R \rfloor + 1} \tag{12.10}$$

satisfies

$$\Upsilon(R) = \mathcal{C}(\lfloor R \rfloor + 1), \qquad \mathcal{C}(R) = \Upsilon(R) - M(R)\Big(\frac{1}{R} - \frac{1}{\lfloor R \rfloor + 1}\Big), \tag{12.11}$$

and

$$\mathcal{C}(R) = -\int_R^\infty \frac{M(w)}{w^2}\, dw, \qquad \Upsilon(R) = -\sum_{n > R} \frac{M(n)}{n(n+1)}. \tag{12.12}$$

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (12.11). For $s > 1$, (8.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s)) \to 0$ as $s \downarrow 1$. For $s \in [1, 1+\delta]$ the bound $M(w) \ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw = 0$. □

Remark 12.11 ($\mathcal{C}$ versus Υ). Summing (12.7) against $M(y)$ suggests the coefficient $\Upsilon(R)$, but the truncated top multiplicity replaces it by $\mathcal{C}(R)$; by (12.11) the two agree exactly when $M(\lfloor R \rfloor) = 0$. For $1 < R < 2$ one has $M_U^{\mathrm{lp}} = -(\pi(t) - \pi(U^2)) \sim -(R-1)U^2/(2\log U)$, i.e. $\mathcal{C}(R) = 1 - 1/R$, whereas $\Upsilon(R) = 1/2$.

Corollary 12.12 (Decay of the coefficients)**.** *By* (12.12) *and* (7.2),

$$\mathcal{C}(R),\ \Upsilon(R) \ll \exp(-c_1(\log R)^{3/5}(\log\log R)^{-1/5}). \tag{12.13}$$

Proposition 12.13 (Both parts are large)**.** (i) *For fixed $R > 1$ with $\mathcal{C}(R) \neq 0$,*

$$M_U^{\mathrm{lp}}(RU^2) \sim -\frac{RU^2}{2\log U}\mathcal{C}(R), \qquad M_U^{\mathrm{sm}}(RU^2) \sim +\frac{RU^2}{2\log U}\mathcal{C}(R); \tag{12.14}$$

for $1 < R < 2$, $\mathcal{C}(R) = 1 - 1/R > 0$.

(ii) *There is $C > 0$ such that, as $t \to \infty$, uniformly for $U^2 < t < U^4$ with $R = t/U^2 \geq \exp(C(\log\log t)^2)$,*

$$M_U^{\mathrm{lp}}(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad M_U^{\mathrm{sm}}(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{12.15}$$

In particular, in both regimes $|M_U^{\mathrm{lp}}(t)|$ and $|M_U^{\mathrm{sm}}(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon < 1/2$ and all large t, and only their sum M can be compared with the scale $t^{1/2+\varepsilon}$.

Proof. We use the classical form [14]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}}), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a \leq y}\frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{12.16}$$

and the series $\sum_{n\geq 1}\mu(n)\log n/n = -1$ converges conditionally. This follows, for example, from the displayed bound on $\widetilde{m}$ and partial summation (and agrees with the limit at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\leq Y}\frac{\mu(a)\log a}{a}\Big) = 1. \tag{12.17}$$

(i) Proposition 12.9, and $M(t) = o(t/\log t)$. For $1 < R < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < U^4$, $R < \sqrt{t}$, so $\log(t/a) \geq \frac{1}{2}\log t$ for $a \leq R$. By (12.5),

$$M_U^{\mathrm{lp}}(t) = -\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\leq R}\mu(a)(\pi(t/a) - \mathrm{li}(t/a)) + M(R)\pi(U^2). \tag{12.18}$$

The second term is $\ll t(1+\log R)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac{1}{4}\log t < \log U < \frac{1}{2}\log t$, so

$$|M(R)|\pi(U^2) \ll Re^{-c\sqrt{\log R}}\frac{U^2}{\log U} \ll e^{-c\sqrt{\log R}}\frac{t}{\log t} = o(t/\log^2 t)$$

once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\leq R}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(R)F(R) - \int_1^R \widetilde{m}(w)F'(w)\,dw, \tag{12.19}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big).$$

The boundary term is $\ll e^{-c\sqrt{\log R}}t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1,4].$$

The required uniformity in (U,R) follows by an explicit ε–W argument. Given $\varepsilon_1 > 0$, choose W so that

$$\int_W^\infty \frac{|\widetilde{m}(w)|}{w}\,dw < \varepsilon_1.$$

For all sufficiently large t, the lower bound on R gives $R > W$. On $[1,W]$,

$$\sup_{1\le w\le W}|g_t(w)-1| \ll \frac{\log W}{\log t} \to 0,$$

while on $[W,R]$ one has $1 \le g_t(w) \le 4$. The omitted interval (R,∞) is also contained in the chosen tail. Therefore

$$\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} = \int_1^\infty \widetilde{m}(w)\frac{dw}{w} + o(1) = 1 + o(1),$$

uniformly over all admissible pairs (U,R). Hence $M_U^{\mathrm{lp}}(t) = (1+o(1))t/\log^2 t$, and $M_U^{\mathrm{sm}} = M - M_U^{\mathrm{lp}}$ with $M(t) = o(t/\log^2 t)$. □

Remark 12.14 (Why $t/\log^2 t$). $\mathcal{C}(R)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 12.12. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and does not decay in R.

Remark 12.15 (First copy transitions). For $1 < R < 2$ every copy has length 1 and $M_U^{\mathrm{lp}}(t) = -(\pi(t) - \pi(U^2))$. For $2 < R < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < R < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

13. Local large-prime decomposition of an excursion

Proposition 13.1 (Local decomposition). *Let $I_\nu = (u,v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}} \mu(n) - \sum_{a<y}\mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y,\tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{13.1}$$

and the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}}\mu(a)\,\Pi_a(v). \tag{13.2}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a,p) \mapsto ap$ is a bijection onto these n. □

Remark 13.2 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Huxley proved the prime number theorem uniformly in intervals of length $h \ge x^{7/12+\varepsilon}$ [7]; the zero-density estimates of Guth–Maynard give intervals of length $x^{17/30+o(1)}$ [3]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta}\mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [13]. The cited results do not provide, below these ranges, an asymptotic for Π_a uniform in a.

Lemma 13.3 (Inverse logarithms)**.** *Let C be as in Proposition 12.13. Uniformly for*

$$\exp(C(\log\log x)^2) \le z \le \sqrt{x},$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{13.3}$$

Proof. Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw.$$

Since $z \le \sqrt{x}$, the boundary term is uniformly $O(e^{-c\sqrt{\log z}}/\log x)$, hence $o(1/\log^2 x)$. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$ with $1 \le g_x(w) \le 4$, extend $g_x\mathbf{1}_{\{w\le z\}}$ by zero, and apply dominated convergence with (12.17). □

Heuristic 13.4 (Opposite drifts)**.** Let $u \ge \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}}\mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 13.3. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of σ_ν. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu,$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (13.2) is a balance between two opposite drifts: the local analogue of Proposition 12.13.

14. A one-frequency model for the black curve

Heuristic 14.1 (Scale and period)**.** Motivated by (6.1), model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, and retain only the lowest zero γ_1 (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so the black curve oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{14.1}$$

the scale of Corollary 7.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in x. In this model consecutive zeros of M are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$, the threshold of Corollary 9.13(iii): a model excursion is exactly λ_*-long.

15. Numerical data

The following census covers $n \le N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions, recording $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) = 6.3449\times10^9, \quad \mathcal{A}_-(N) = -7.0877\times10^9, \quad \mathcal{A}(N) = -7.4279\times10^8, \tag{15.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{15.2}$$

Since $|M(x)| \le 0.571\sqrt{x}$ for $33 \le x \le 10^{16}$ [6], and $|M(n)| \le \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $C_0 = 1$ is admissible in Lemma 7.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \le \frac{4}{5}N^{5/4} = 5.317\times10^{10}$ against the observed 1.3433×10^{10}, a ratio of 3.96. Heuristic 8.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants. Reproducing these values requires the excursion file, the area-integration convention of Proposition 4.2, and the terminal convention of Definition 3.6. These data are not used in any proof.

16. Concluding remarks

The main results can be summarized in five statements. First, the Kaczorowski–Pintz oscillation density, combined with elementary counts of lacunary families of excursions, localizes Littlewood's criterion to the initial part $x_0 < t \le \min\{x_1, \lambda x_0\}$ of each excursion for every $\lambda > \lambda_* = e^{\pi/\gamma_1}$, the sharp threshold of the counting mechanism; in averaged form, the excursions carrying large values have mean logarithmic length at most $\log \lambda_*$. Second, on this window the future increments depend directly only on the Möbius prefix up to θu for every $\theta > \theta_* = \frac{1}{2}e^{\pi/\gamma_1}$, and the endpoints of late excursions satisfy $x_1 = x_0^{1+o(1)}$. Third, the remaining difficulty is not one of dependency: the one-step recursion, the relaxed divisor-sum cone and any bounded number of multiplicative layers provably fail to produce the RH scale by themselves. Fourth, Pintz's 2026 oscillation theorem transfers to the excursion variables as exact logarithmic exponent identities: cumulative absolute area has exponent $1 + \Theta$, quadratic mass has exponent $1 + 2\Theta$, and the chord-energy functional has exponent $1 + \Theta$. The same input makes Θ exactly the abscissa of absolute convergence of the completed-excursion block series for $1/\zeta$. Fifth, in the language of two-point correlations, RH is equivalent to cancellation between lags of the h-weighted correlations at the zeros of M. More precisely, the total weighted mass has exponent $1+2\Theta$, and every range $h \le v^\alpha$, $\alpha < \Theta$, is negligible relative to it; the universal endpoint statement $h \le \sqrt{v}$ remains $o(v^2)$. Uniform absolute two-point Cesàro cancellation is available in every interval of length $\ge u^{3/5+\eta}$ down to lags $R \ge L(\log u)^{-2+\delta}$. A distinct, weaker Fejér target on a complete excursion is equivalent, for $R = o(\sqrt{L})$, to a diagonal local-variance law for increments of M. Absolute Chowla, this Fejér target, and localization of the weighted energy are separate problems and are not claimed to be equivalent.

17. Open problems

(1) Extend the absolute two-point estimate of Theorem 10.13 to every adaptively selected excursion with $x_0^{1/2+\varepsilon} \le \ell < x_0^{3/5}$, and if possible to lags much smaller than $\ell(\log x_0)^{-2+\delta}$. This is a stronger target than the Fejér problem below.

(2) For complete excursions with $L \ge u^{1/2+\varepsilon}$, prove for some $R \to \infty$, $R = o(\sqrt{L})$, the diagonal variance law
$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = \left(\frac{6}{\pi^2} + o(1)\right) RL.$$
By Proposition 10.20 this is equivalent to $F_R = \sum_{h<R}(R-h)C_h = o(RL)$, but not to absolute averaged Chowla and not, without further information, to $\sum_{h<R} hC_h = o(Q)$.

(3) In the localized form of Corollary 9.52, bound
$$\sum_{d \le 5u/8} \mu(d)\big(\lfloor t/d \rfloor - \lfloor u/d \rfloor\big)$$
by $t^{1/2+\varepsilon}$ on $u < t \le \min\{x_1, 5u/4\}$, using the full simultaneous intra-excursion constraints rather than pointwise estimates.

(4) Quantify the multiscale persistence forced by Lemma 9.45. By Corollary 9.47 and Remark 9.48, channel descent iterates but loses the reciprocal factor $2K/k^\alpha > 1$ per step. Since the guaranteed normalized factor $k^\alpha/(2K)$ is always $< 1/2$, the loss cannot disappear at a single step within the present pigeonhole argument. The problem is to recover it collectively from the persistent cluster forced by Lemma 9.45, or to establish genuine cancellation for the deep remainder $\mathcal{R}_K$.

(5) Question 9.65 in the form left by Corollary 9.67: bounded depth fixes only $o(u)$ window increments, whose sum is of order $u(\log\log u)^k/\log u$; decide whether the unit-step and excursion constraints on the remaining free coordinates δ_d, $\Omega(d) > k$, already force the cancellation down to $u^{1/2+\varepsilon}$. The first case is $k = 1$ (primes fixed, $\delta_p = -1$).

(6) Can the threshold $\lambda_* = e^{\pi/\gamma_1}$ of Corollary 9.13 be lowered? By Remark 9.29 this requires information on the position of large values inside excursions, beyond the count (9.6).
(7) Whether (7.4) is equivalent to RH.
(8) Whether the pruned local condition (8.7) holds, and whether the bridge constraints (5.5), (5.6) admit a third independent relation involving the defect $\ell_\nu/2 - H_\nu \geq 0$.
(9) Establish (SG) for $Z = H/\sqrt{\ell}$ in the simple random walk model of the Möbius walk, and compute the extreme scale of Corollary 9.71 there; the Brownian-excursion limit suggests $c = 2$.
(10) Determine what happens at the endpoint of Theorem 10.6: is the contribution of $h \leq v^{\Theta}$ negligible relative to $\sum_{n<v} M(n)^2$ along zeros v? Even under RH the theorem presently gives relative negligibility only for $h \leq v^{1/2-\delta}$, while Proposition 10.4 gives the endpoint only at the absolute v^2 scale.
(11) Determine convergence on the boundary $\Re s = \Theta$ for the excursion-block series of Theorem 9.78. The abscissa is exact, but the argument does not decide boundary convergence.
(12) Numerically: the number of excursions with $\ell_\nu > \sqrt{N}$ against $N^{1/4}$, the maxima of $H_\nu/\sqrt{\ell_\nu}$ and $\beta_\nu/\ell_\nu^{3/2}$ over them; the split (13.1) and the σ_ν-independence of Heuristic 13.4; the frequency of touching runs and of ballistic excursions (Heuristic 8.8); and the distribution of x_1/x_0, to be compared with λ_*.

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